

Chapter 6: Bessel Functions

Special Functions of Mathematical Physics

after K. A. Novak
Compiled by Jake W. Liu

July 25, 2026

Outline

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation
- 4 Asymptotics
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series
- 7 Summary

Where we are going

Bessel's equation of integer order m is

$$x^2 y''(x) + xy'(x) + (x^2 - m^2)y(x) = 0. \quad (6.0.1)$$

It arises when variables are separated in circular or cylindrical geometries. This chapter will establish, with all index shifts and normalization factors shown explicitly,

- a generating function for the integer-order family J_m , and the power series;
- the order and derivative recurrences;
- one carefully derived real integral representation (plus a brief contour form);
- the small- and large-argument expansions (large x by stationary phase, the real oscillatory case of steepest descent from Ch. 2);
- the second solution Y_m , a short look at Hankel functions $H^{(1)}, H^{(2)}$, and the discrete Fourier–Bessel series on the disk.

Throughout the generating-function sections, $m, n \in \mathbb{Z}$ unless stated otherwise. For small- or large-argument limits, the order is held fixed. On the infinite half-line the discrete Fourier–Bessel sum becomes a Hankel integral transform; that continuous transform is not developed here.

Table of contents

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation
- 4 Asymptotics
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series
- 7 Summary

Generating function for the family $\{J_m\}$ I

Why a generating function? Separation of variables in polar or cylindrical coordinates produces an infinite family of radial solutions, one for each integer angular order $m \in \mathbb{Z}$. Writing a separate series for every m works, but it is wasteful: recurrences that link neighboring orders, and integral representations that extract a single order, are far cleaner if the whole family is packaged into one closed object $g(x, t)$.

A standard packaging for a bi-infinite sequence $\{P_m(x)\}_{m \in \mathbb{Z}}$ is a *Laurent generating function*

$$g(x, t) = \sum_{m=-\infty}^{\infty} P_m(x) t^m, \quad t \neq 0. \quad (6.1.1)$$

One closed formula for g then yields every P_m by coefficient extraction, and differentiating g with respect to t or x produces the recurrence identities automatically.

Which closed form? For integer-order Bessel functions we adopt the classical choice

$$g(x, t) = \exp\left[\frac{x}{2} (t - t^{-1})\right] = \sum_{m=-\infty}^{\infty} J_m(x) t^m, \quad t \neq 0, \quad (6.1.2)$$

Generating function for the family $\{J_m\}$ II

and *define* $J_m(x)$ as the Laurent coefficient of t^m . This is not pulled from the sky: the combination $t - t^{-1}$ is precisely what appears when a plane wave $e^{ix \sin \theta}$ is expanded in angular Fourier modes (set $t = e^{i\theta}$ later). Once (6.1.2) is in place we will

- 1 extract the power series of J_m by the Cauchy product of two exponentials;
- 2 verify term by term that the series solves Bessel's equation (6.0.1);
- 3 derive recurrences and one real integral representation from the same g .

Two-minute review: Laurent coefficient extraction I

Step 1: ordinary power series first. If $f(t) = \sum_{n=0}^{\infty} a_n t^n$ is analytic at the origin, then

$$a_n = [t^n] f(t) = \frac{f^{(n)}(0)}{n!}.$$

The same idea works for series with both positive and negative powers.

Step 2: Laurent series on an annulus. A function analytic for $0 < r < |t| < R$ admits a unique expansion

$$f(t) = \sum_{m=-\infty}^{\infty} c_m t^m, \quad c_m = [t^m] f(t).$$

On any positively oriented simple closed contour C lying in that annulus and winding once about 0, Cauchy's integral formula extracts the coefficient:

$$[t^m] f(t) = \frac{1}{2\pi i} \oint_C f(t) t^{-m-1} dt. \quad (6.1.3)$$

Special cases used constantly below:

- if $f(t) = \sum_m c_m t^m$, then $[t^m] f = c_m$ by uniqueness;
- for an integer power, $[t^m] t^k = \delta_{mk}$ (Kronecker delta);

Two-minute review: Laurent coefficient extraction II

- the product of two Laurent series has coefficients given by the Cauchy convolution: if $f = \sum a_j t^j$ and $g = \sum b_k t^k$, then $[t^m](fg) = \sum_{j+k=m} a_j b_k$.

Step 3: what we will do with $g(x, t)$. Write $g(x, t) = e^{xt/2} e^{-x/(2t)}$, expand each exponential as a power series, multiply, and collect the coefficient of t^m . By definition that coefficient is $J_m(x)$.

Reading off the integer-order series I

Step 1: expand both exponentials. For fixed x , the product representation

$$g(x, t) = e^{xt/2} e^{-x/(2t)}$$

converges locally uniformly on every compact annulus $0 < r \leq |t| \leq R < \infty$: on $|t| \leq R$ the series $e^{xt/2} = \sum_{j \geq 0} \frac{(x/2)^j}{j!} t^j$ converges uniformly, and on $|t| \geq r > 0$ the series $e^{-x/(2t)} = \sum_{k \geq 0} \frac{(-1)^k (x/2)^k}{k!} t^{-k}$ converges uniformly. Both are absolutely convergent, so their Cauchy product is an absolutely and locally uniformly convergent Laurent series. Absolute convergence lets us regroup by the power of t and differentiate termwise in x or t on such annuli.

Explicitly,

$$\begin{aligned} e^{xt/2} &= \sum_{j=0}^{\infty} \frac{1}{j!} \left(\frac{xt}{2}\right)^j = \sum_{j=0}^{\infty} \frac{1}{j!} \left(\frac{x}{2}\right)^j t^j, \\ e^{-x/(2t)} &= \sum_{k=0}^{\infty} \frac{1}{k!} \left(-\frac{x}{2t}\right)^k = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \left(\frac{x}{2}\right)^k t^{-k}. \end{aligned}$$

Reading off the integer-order series II

Multiplying gives the double series

$$\begin{aligned} g(x, t) &= \left(\sum_{j=0}^{\infty} \frac{(x/2)^j}{j!} t^j \right) \left(\sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^k}{k!} t^{-k} \right) \\ &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k}{j! k!} \left(\frac{x}{2} \right)^{j+k} t^{j-k}. \end{aligned}$$

Step 2: impose the power constraint. To extract the coefficient of t^m for $m \geq 0$, impose

$$j - k = m.$$

Since $j, k \geq 0$, this is equivalent to

$$j = k + m, \quad k = 0, 1, 2, \dots$$

Then

$$j + k = (k + m) + k = 2k + m, \quad j! = (k + m)!.$$

Therefore

$$[t^m]g(x, t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k + m)! k!} \left(\frac{x}{2} \right)^{2k+m}.$$

Reading off the integer-order series III

By (6.1.2), $[t^m]g = J_m$, so

$$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(m+k)!} \left(\frac{x}{2}\right)^{2k+m}, \quad m = 0, 1, 2, \dots \quad (6.1.4)$$

The first few series are

$$\begin{aligned} J_0(x) &= 1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \dots, \\ J_1(x) &= \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} - \frac{x^7}{18432} + \dots, \\ J_2(x) &= \frac{x^2}{8} - \frac{x^4}{96} + \frac{x^6}{3072} + \dots. \end{aligned}$$

The generating function contains only integer powers of t ; therefore it generates integer-order functions J_m . The general-order series J_ν is stated next and is not a Laurent coefficient when $\nu \notin \mathbb{Z}$.

General order and direct verification of Bessel's equation I

Step 1: state the general-order series. For a general order ν , with Γ denoting the gamma function, the Frobenius series is

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}. \quad (6.1.5)$$

Where this series comes from. It is not a new postulate: (6.1.5) is the $r = \nu$ Frobenius solution built in Chapter 4. Substituting $y = \sum_{k \geq 0} a_k x^{k+\nu}$ into Bessel's equation and using $(k + \nu)^2 - \nu^2 = k(k + 2\nu)$ collapses the equation to the two-term recurrence

$$k(k + 2\nu)a_k + a_{k-2} = 0, \quad a_{-1} = a_{-2} = 0,$$

which links a_k only to a_{k-2} . Taking $a_1 = 0$ therefore removes the odd chain, while the even chain $a_{2j} = -a_{2j-2}/(4j(j + \nu))$ unwinds to

$$a_{2j} = (-1)^j \frac{\Gamma(\nu + 1)}{2^{2j} j! \Gamma(j + \nu + 1)} a_0,$$

where the product $\prod_{\ell=1}^j (\nu + \ell) = \Gamma(\nu + j + 1)/\Gamma(\nu + 1)$ was used; that product formula is just $\Gamma(z + 1) = z \Gamma(z)$ (Chapter 3) applied j times. The overall scale a_0 is free, and the conventional choice

$$a_0 = \frac{2^{-\nu}}{\Gamma(\nu + 1)}$$

General order and direct verification of Bessel's equation II

cancels the factor $\Gamma(\nu + 1)$ and leaves $a_{2j}x^{2j+\nu} = \frac{(-1)^j}{j! \Gamma(j+\nu+1)} (x/2)^{2j+\nu}$, which is exactly (6.1.5). This normalization is chosen so that the general-order series agrees with the integer-order series read off the generating function, as the next paragraph checks. For $x > 0$, $(x/2)^\nu = e^{\nu \log(x/2)}$ is unambiguous. For complex x , a branch of $\log x$ must be fixed when $\nu \notin \mathbb{Z}$. If $\nu = m \geq 0$ is an integer, then $\Gamma(k + m + 1) = (k + m)!$, and (6.1.5) reduces to (6.1.4).

For fixed ν and fixed branch, let $T_k(x)$ denote the k -th term of (6.1.5). For all sufficiently large k ,

$$\left| \frac{T_{k+1}(x)}{T_k(x)} \right| = \frac{|x|^2}{4(k+1)|k+\nu+1|} \rightarrow 0.$$

The ratio tends to zero uniformly for x in a compact set. The differentiated series have this same quadratic decay in k , up to polynomial factors. Thus the series and the series obtained by differentiating once or twice with respect to x converge locally uniformly away from the chosen branch cut. This justifies the termwise substitutions below.

General order and direct verification of Bessel's equation III

Step 2: verify the differential equation term by term. We now verify directly that (6.1.4) satisfies Bessel's equation. Write

$$J_m(x) = \sum_{k=0}^{\infty} a_k x^{2k+m}, \quad a_k = \frac{(-1)^k}{2^{2k+m} k! (m+k)!}.$$

For one monomial $a_k x^{2k+m}$,

$$x^2 \frac{d^2}{dx^2} (a_k x^{2k+m}) = a_k (2k+m)(2k+m-1) x^{2k+m},$$

$$x \frac{d}{dx} (a_k x^{2k+m}) = a_k (2k+m) x^{2k+m}.$$

Adding these two expressions and subtracting $m^2 a_k x^{2k+m}$ gives

$$\begin{aligned} & [(2k+m)(2k+m-1) + (2k+m) - m^2] a_k x^{2k+m} \\ &= [(2k+m)^2 - m^2] a_k x^{2k+m} \\ &= [(2k+m-m)(2k+m+m)] a_k x^{2k+m} \\ &= 4k(k+m) a_k x^{2k+m}. \end{aligned}$$

General order and direct verification of Bessel's equation IV

Hence

$$\begin{aligned} x^2 J_m'' + x J_m' - m^2 J_m \\ = \sum_{k=0}^{\infty} 4k(k+m) a_k x^{2k+m} = \sum_{k=1}^{\infty} 4k(k+m) a_k x^{2k+m}, \end{aligned}$$

because the $k = 0$ term is zero. The remaining term in Bessel's equation is

$$\begin{aligned} x^2 J_m(x) &= \sum_{k=0}^{\infty} a_k x^{2k+m+2} \\ &= \sum_{k=1}^{\infty} a_{k-1} x^{2k+m}, \end{aligned}$$

General order and direct verification of Bessel's equation V

where the second line uses the index shift $k \mapsto k + 1$ in reverse: writing the series as $\sum_{k=0}^{\infty} a_k x^{2k+m+2}$ and renaming the summation index $k' = k + 1$ (so when $k = 0$, $k' = 1$, and $a_k = a_{k'-1}$), then dropping the prime, yields $\sum_{k=1}^{\infty} a_{k-1} x^{2k+m}$. Finally,

$$\begin{aligned} a_{k-1} &= \frac{(-1)^{k-1}}{2^{2k+m-2}(k-1)!(m+k-1)!} \\ &= -\frac{4k(m+k)(-1)^k}{2^{2k+m}k!(m+k)!} \\ &= -4k(m+k)a_k. \end{aligned}$$

Check the factor of $4k(m+k)$ carefully:

$$\begin{aligned} \frac{a_{k-1}}{a_k} &= \frac{(-1)^{k-1}}{(-1)^k} \cdot \frac{2^{2k+m}}{2^{2k+m-2}} \cdot \frac{k!}{(k-1)!} \cdot \frac{(m+k)!}{(m+k-1)!} \\ &= (-1) \cdot 2^2 \cdot k \cdot (m+k) = -4k(m+k). \end{aligned}$$

Thus every coefficient in

$$x^2 J_m'' + x J_m' + (x^2 - m^2) J_m = \sum_{k=1}^{\infty} [4k(k+m)a_k + a_{k-1}] x^{2k+m}$$

vanishes, proving

$$x^2 J_m''(x) + x J_m'(x) + (x^2 - m^2) J_m(x) = 0. \quad (6.1.6)$$

General order and direct verification of Bessel's equation VI

The same coefficient calculation verifies the general-order series. With

$$a_k(\nu) = \frac{(-1)^k}{2^{2k+\nu} k! \Gamma(k + \nu + 1)},$$

the gamma recurrence $\Gamma(k + \nu + 1) = (k + \nu)\Gamma(k + \nu)$ gives

$$\begin{aligned} a_{k-1}(\nu) &= \frac{(-1)^{k-1}}{2^{2k+\nu-2} (k-1)! \Gamma(k + \nu)} \\ &= -4k(k + \nu)a_k(\nu). \end{aligned}$$

At exceptional values for which a gamma factor has a pole, the same identity follows by continuity because $1/\Gamma(z)$ is entire and vanishes at the nonpositive integers.

Replacing m by ν in the preceding coefficient identity therefore proves

$$x^2 J_\nu''(x) + x J_\nu'(x) + (x^2 - \nu^2) J_\nu(x) = 0$$

for the chosen branch of the series.

Negative integer orders: $J_{-m} = (-1)^m J_m$

Step 1: read the negative coefficient off g itself. By (6.1.2) the symbol J_{-m} means the Laurent coefficient $[t^{-m}]g$, so the honest starting point is the double series found earlier,

$$g(x, t) = \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k}{j! k!} \left(\frac{x}{2}\right)^{j+k} t^{j-k}.$$

Fix $m \in \mathbb{Z}_{\geq 0}$ and impose $j - k = -m$. Since $j, k \geq 0$, this is equivalent to

$$k = j + m, \quad j = 0, 1, 2, \dots,$$

the mirror image of the constraint $j = k + m$ used for nonnegative orders. Then $j + k = j + (j + m) = 2j + m$, $k! = (j + m)!$, $(-1)^k = (-1)^{j+m} = (-1)^m (-1)^j$,
so

$$\begin{aligned} J_{-m}(x) &= [t^{-m}]g(x, t) = \sum_{j=0}^{\infty} \frac{(-1)^{j+m}}{j! (j+m)!} \left(\frac{x}{2}\right)^{2j+m} \\ &= (-1)^m \sum_{j=0}^{\infty} \frac{(-1)^j}{j! (m+j)!} \left(\frac{x}{2}\right)^{2j+m} \\ &= (-1)^m J_m(x), \end{aligned}$$

Negative integer orders: $J_{-m} = (-1)^m J_m$ II

the last equality by (6.1.4). This settles (6.1.7) for the definition actually in force, and it closes a second loop as well: the Laurent coefficient $[t^{-m}]g$ is now known, so comparing it with the computation below shows that the general-order series (6.1.5) evaluated at $\nu = -m$ returns the same function. From here on the two names for J_{-m} may be used interchangeably.

Step 2: the same result from the general-order series.

Let $m \in \mathbb{Z}_{\geq 0}$. For $x \neq 0$, substitute $\nu = -m$ into the general-order series (6.1.5):

$$J_{-m}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k - m + 1)} \left(\frac{x}{2}\right)^{2k-m}.$$

For $k = 0, 1, \dots, m-1$, the argument $k - m + 1$ is a nonpositive integer. The gamma function has a pole there, so its reciprocal is zero:

$$\frac{1}{\Gamma(k - m + 1)} = 0, \quad 0 \leq k \leq m-1.$$

Consequently the first nonzero term is $k = m$:

$$J_{-m}(x) = \sum_{k=m}^{\infty} \frac{(-1)^k}{k! (k - m)!} \left(\frac{x}{2}\right)^{2k-m}.$$

Negative integer orders: $J_{-m} = (-1)^m J_m$ III

Set $k = \ell + m$. Then $\ell = 0, 1, 2, \dots$, and

$$(-1)^k = (-1)^{\ell+m} = (-1)^m (-1)^\ell, \quad 2k - m = 2(\ell + m) - m = 2\ell + m.$$

Also $k! = (\ell + m)!$ and $(k - m)! = \ell!$. Therefore

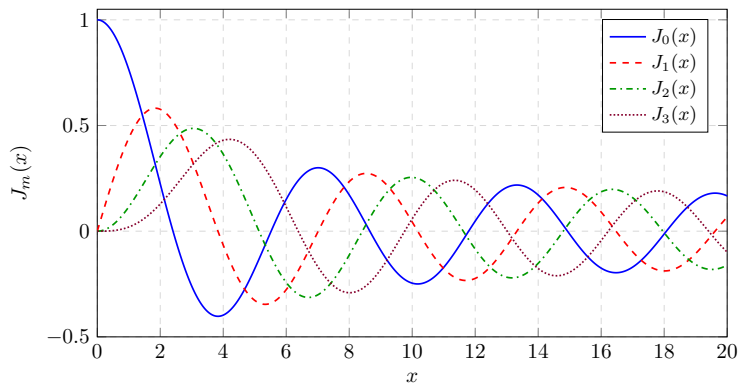
$$\begin{aligned} J_{-m}(x) &= \sum_{\ell=0}^{\infty} \frac{(-1)^{\ell+m}}{(\ell + m)! \ell!} \left(\frac{x}{2}\right)^{2\ell+m} \\ &= (-1)^m \sum_{\ell=0}^{\infty} \frac{(-1)^\ell}{\ell! (\ell + m)!} \left(\frac{x}{2}\right)^{2\ell+m} \\ &= (-1)^m J_m(x). \end{aligned}$$

Thus, first for $x \neq 0$ and then at $x = 0$ by continuity,

$$J_{-m}(x) = (-1)^m J_m(x), \quad m \in \mathbb{Z}. \quad (6.1.7)$$

In particular, $J_{-1} = -J_1$, $J_{-2} = J_2$, and negative integer orders do not provide a second linearly independent solution.

Bessel functions of the first kind



For $m \geq 0$, the leading term in (6.1.4) is $J_m(x) = (x/2)^m/m! + O(x^{m+2})$. Thus $J_0(0) = 1$ and $J_m(0) = 0$ for $m \geq 1$.

Jacobi–Anger: the generating function on $|t| = 1$

Set $t = e^{i\theta}$ in (6.1.2). Since $t^{-1} = e^{-i\theta}$,

$$\begin{aligned}t - t^{-1} &= e^{i\theta} - e^{-i\theta} \\&= (\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta) \\&= 2i \sin \theta.\end{aligned}$$

Therefore

$$\begin{aligned}\exp\left[\frac{x}{2}(t - t^{-1})\right] &= \exp\left[\frac{x}{2}(2i \sin \theta)\right] \\&= e^{ix \sin \theta}.\end{aligned}$$

On the series side, $t^m = e^{im\theta}$. Hence

$$e^{ix \sin \theta} = \sum_{m=-\infty}^{\infty} J_m(x) e^{im\theta}. \quad (6.1.8)$$

This is the Jacobi–Anger expansion: a plane wave in polar phase is the Fourier series of the integer-order Bessel family. For fixed x the series $\sum_m J_m(x) e^{im\theta}$ is the restriction of the Laurent series (6.1.2) to the circle $|t| = 1$, so it converges uniformly in $\theta \in [0, 2\pi]$, and a uniformly convergent series may be integrated term by term.

Jacobi–Anger: the generating function on $|t| = 1$ II

Multiplying by $e^{-in\theta}$ and integrating over one period therefore gives the coefficient explicitly:

$$\frac{1}{2\pi} \int_0^{2\pi} e^{ix \sin \theta} e^{-in\theta} d\theta = \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} J_m(x) \int_0^{2\pi} e^{i(m-n)\theta} d\theta.$$

For an integer q ,

$$\int_0^{2\pi} e^{iq\theta} d\theta = \int_0^{2\pi} 1 d\theta = 2\pi, \quad q = 0,$$

$$\int_0^{2\pi} e^{iq\theta} d\theta = \left. \frac{e^{iq\theta}}{iq} \right|_0^{2\pi} = \frac{e^{i2\pi q} - 1}{iq} = 0, \quad q \neq 0.$$

Equivalently, $\int_0^{2\pi} e^{iq\theta} d\theta = 2\pi \delta_{q0}$. Thus only $m = n$ survives:

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta, \quad n \in \mathbb{Z}. \quad (6.1.9)$$

This complex integral is the starting point for the real trigonometric form derived next.

Table of contents

- 1 Generating function
- 2 Recursion formulae**
- 3 Integral representation
- 4 Asymptotics
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series
- 7 Summary

Differentiating in t : the three-term recurrence I

Step 1: differentiate the closed form. Differentiate the closed form in (6.1.2). The exponent is

$$h(t) = \frac{x}{2} (t - t^{-1}),$$

so

$$\begin{aligned} h'(t) &= \frac{x}{2} \left(\frac{d}{dt} t - \frac{d}{dt} t^{-1} \right) \\ &= \frac{x}{2} (1 - (-t^{-2})) \\ &= \frac{x}{2} (1 + t^{-2}). \end{aligned}$$

By the chain rule,

$$\begin{aligned} \frac{\partial g}{\partial t} &= h'(t) e^{h(t)} \\ &= \frac{x}{2} (1 + t^{-2}) g(x, t). \end{aligned}$$

Differentiating in t : the three-term recurrence II

Insert the Laurent series for g :

$$\begin{aligned}\frac{\partial g}{\partial t} &= \frac{x}{2}(1+t^{-2}) \sum_{m=-\infty}^{\infty} J_m(x)t^m \\ &= \frac{x}{2} \sum_{m=-\infty}^{\infty} J_m(x)t^m + \frac{x}{2} \sum_{m=-\infty}^{\infty} J_m(x)t^{m-2}.\end{aligned}$$

In the second sum, set $m = r + 2$, equivalently $r = m - 2$:

$$\sum_{m=-\infty}^{\infty} J_m(x)t^{m-2} = \sum_{r=-\infty}^{\infty} J_{r+2}(x)t^r.$$

Renaming r as m gives

$$\frac{\partial g}{\partial t} = \frac{x}{2} \sum_{m=-\infty}^{\infty} [J_m(x) + J_{m+2}(x)]t^m. \quad (6.2.1)$$

Step 2: differentiate the Laurent series. Differentiate the Laurent series directly:

$$\frac{\partial g}{\partial t} = \sum_{m=-\infty}^{\infty} mJ_m(x)t^{m-1}.$$

Differentiating in t : the three-term recurrence III

Set $m = r + 1$, so that $m - 1 = r$:

$$\begin{aligned}\frac{\partial g}{\partial t} &= \sum_{r=-\infty}^{\infty} (r+1)J_{r+1}(x)t^r \\ &= \sum_{m=-\infty}^{\infty} (m+1)J_{m+1}(x)t^m.\end{aligned}$$

Equating the coefficients of t^m in this expression and (6.2.1) yields

$$(m+1)J_{m+1}(x) = \frac{x}{2} [J_m(x) + J_{m+2}(x)].$$

Replace m by $m - 1$:

$$mJ_m(x) = \frac{x}{2} [J_{m-1}(x) + J_{m+1}(x)].$$

For $x \neq 0$, multiplication by $2/x$ gives

$$J_{m-1}(x) + J_{m+1}(x) = \frac{2m}{x} J_m(x). \quad (6.2.2)$$

At $x = 0$, any endpoint value is obtained by taking a limit from the power series rather than by substituting into the quotient $2mJ_m/x$.

Differentiating in x : the derivative recurrence I

Differentiate the closed form in (6.1.2) with respect to x . Since

$$\frac{\partial}{\partial x} \left[\frac{x}{2}(t - t^{-1}) \right] = \frac{1}{2}(t - t^{-1}),$$

the chain rule gives

$$\begin{aligned} \frac{\partial g}{\partial x} &= \frac{1}{2}(t - t^{-1})g(x, t) \\ &= \frac{1}{2}(t - t^{-1}) \sum_{m=-\infty}^{\infty} J_m(x)t^m \\ &= \frac{1}{2} \sum_{m=-\infty}^{\infty} J_m(x)t^{m+1} - \frac{1}{2} \sum_{m=-\infty}^{\infty} J_m(x)t^{m-1}. \end{aligned}$$

For the first sum, set $m = r - 1$; for the second, set $m = r + 1$:

$$\begin{aligned} \sum_m J_m t^{m+1} &= \sum_r J_{r-1} t^r, \\ \sum_m J_m t^{m-1} &= \sum_r J_{r+1} t^r. \end{aligned}$$

Differentiating in x : the derivative recurrence II

After renaming r as m ,

$$\frac{\partial g}{\partial x} = \frac{1}{2} \sum_{m=-\infty}^{\infty} [J_{m-1}(x) - J_{m+1}(x)] t^m. \quad (6.2.3)$$

On the other hand, direct differentiation of the series gives

$$\frac{\partial g}{\partial x} = \sum_{m=-\infty}^{\infty} J'_m(x) t^m.$$

Equating coefficients of t^m gives

$$2J'_m(x) = J_{m-1}(x) - J_{m+1}(x). \quad (6.2.4)$$

We can combine (6.2.2) and (6.2.4) to eliminate either neighboring order. First solve (6.2.2) for J_{m+1} :

$$J_{m+1} = \frac{2m}{x} J_m - J_{m-1}.$$

Substitute this into (6.2.4):

$$\begin{aligned} 2J'_m &= J_{m-1} - \left(\frac{2m}{x} J_m - J_{m-1} \right) \\ &= 2J_{m-1} - \frac{2m}{x} J_m. \end{aligned}$$

Differentiating in x : the derivative recurrence III

Dividing by 2 gives

$$J'_m(x) = J_{m-1}(x) - \frac{m}{x} J_m(x), \quad x \neq 0. \quad (6.2.5)$$

Alternatively, solve (6.2.2) for J_{m-1} :

$$J_{m-1} = \frac{2m}{x} J_m - J_{m+1}.$$

Substitution into (6.2.4) gives

$$\begin{aligned} 2J'_m &= \left(\frac{2m}{x} J_m - J_{m+1} \right) - J_{m+1} \\ &= \frac{2m}{x} J_m - 2J_{m+1}, \end{aligned}$$

and therefore

$$J'_m(x) = \frac{m}{x} J_m(x) - J_{m+1}(x), \quad x \neq 0. \quad (6.2.6)$$

These identities are exact algebraic consequences of the generating function: they hold with equality for the true values of J_m . They should not be read as a claim about *computation*. If one uses (6.2.2) in the direction of increasing order, computing $J_{m+1}(x) = \frac{2m}{x} J_m(x) - J_{m-1}(x)$ from the two lower orders (*upward recurrence*), then for fixed x and growing m any rounding error in the starting values J_0, J_1 is amplified rather than damped, so the computed sequence loses accuracy. In

Differentiating in x : the derivative recurrence IV

this numerical sense the upward recurrence is *unstable* for large m . The mechanism is worth spelling out, because it is the same for every three-term recurrence.

Read (6.2.2) as a linear difference equation in the order index,

$$u_{m+1} = \frac{2m}{x} u_m - u_{m-1}, \quad (6.2.7)$$

one solution of which is $u_m = J_m(x)$. Prescribing u_0 and u_1 determines the whole sequence, so the solution space is two-dimensional. Because (6.2.7) is linear, the deviation $e_m := \tilde{J}_m - J_m$ of the computed sequence from the exact one obeys that same equation, apart from the fresh rounding committed at each step; so what happens to e_m is decided by how solutions of (6.2.7) behave as m grows. The two possible behaviors are opposite.

On one side stands the Bessel solution itself. Its $k=0$ term in (6.1.4) is $(x/2)^m/m!$, and at fixed x that factorial denominator makes $J_m(x)$ decay faster than any geometric sequence in m . On the other side, suppose u_m is a solution whose magnitude *increases* with m . Once $m > x$ the multiplier $2m/x$ in (6.2.7) exceeds 1 and keeps growing, so the subtracted term u_{m-1} is small beside $(2m/x)u_m$ and $u_{m+1} \approx (2m/x)u_m$. Iterating this from some index M onward,

$$u_m \approx u_M \prod_{k=M}^{m-1} \frac{2k}{x} = u_M \left(\frac{2}{x}\right)^{m-M} \frac{(m-1)!}{(M-1)!} = C \Gamma(m) \left(\frac{2}{x}\right)^m, \quad C = \frac{u_M}{(M-1)!} \left(\frac{x}{2}\right)$$

Differentiating in x : the derivative recurrence V

using $\prod_{k=M}^{m-1} k = (m-1)!/(M-1)!$ and $\Gamma(m) = (m-1)!$. Such a growing solution really is present in the solution space: it is the second Bessel solution Y_m constructed in (6.5.4) below, whose size (6.5.8) carries exactly the factor $\Gamma(m)(2/x)^m$.

Hence e_m is a combination of a factorially decaying and a factorially growing sequence. Exact starting data would put no weight on the growing one; a rounding error of relative size ε in J_0, J_1 puts weight $O(\varepsilon)$ on it, and then, up to a constant,

$$\frac{|e_m|}{|J_m(x)|} \sim |\varepsilon| m! \Gamma(m) \left(\frac{2}{x}\right)^{2m} \rightarrow \infty.$$

At $x = 1$ the amplification factor $m! \Gamma(m) 2^{2m}$ already exceeds 10^{18} at $m = 10$, so seeds carrying the double-precision relative error $\varepsilon \approx 10^{-16}$ leave nothing behind: started from correctly rounded $J_0(1), J_1(1)$, the upward recurrence returns -6.91×10^{-9} for $J_{10}(1)$, whose true value is $+2.63 \times 10^{-10}$ — wrong sign, not one correct digit.

Running the recurrence in the opposite direction,

$$u_{m-1} = \frac{2m}{x} u_m - u_{m+1},$$

interchanges the two roles: now it is the $\Gamma(m)(2/x)^m$ solution that is divided by the large factors at each step while the J component grows relative to it, so an arbitrary starting vector is progressively purged of everything but J_m . The practical obstacle is

Differentiating in x : the derivative recurrence VI

that the two seeds at the top are exactly what one does not know. One therefore starts at some $M \gg m$ with the arbitrary trial values $u_{M+1} = 0$, $u_M = 1$, recurs down to $m = 0$, and fixes the one remaining unknown — the overall scale — at the end. A normalization is already available: setting $\theta = 0$ in (6.1.8) gives $1 = \sum_{m \in \mathbb{Z}} J_m(x)$, and pairing the term m with the term $-m$ through (6.1.7) gives

$$1 = J_0(x) + \sum_{m=1}^{\infty} [1 + (-1)^m] J_m(x) = J_0(x) + 2 \sum_{k=1}^{\infty} J_{2k}(x),$$

since $1 + (-1)^m$ vanishes for odd m and equals 2 for even m . The downward sweep produces $u_m = \lambda J_m(x)$, to within the accuracy sought, for some unknown λ , so $u_0 + 2 \sum_{k \geq 1} u_{2k} = \lambda$, and dividing the computed sequence by that number restores the correct scale. This is how accurate high-order values are obtained in practice.

Consequences and low-order checks I

For $m = 0$, (6.2.6) gives

$$J_0'(x) = -J_1(x). \quad (6.2.8)$$

This agrees term by term with the series:

$$\begin{aligned} J_0'(x) &= \frac{d}{dx} \left(1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \cdots \right) \\ &= -\frac{x}{2} + \frac{x^3}{16} - \frac{x^5}{384} + \cdots \\ &= -\left(\frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} - \cdots \right) \\ &= -J_1(x). \end{aligned}$$

For $m = 1$, (6.2.2) gives

$$\begin{aligned} J_0(x) + J_2(x) &= \frac{2}{x} J_1(x), \\ J_2(x) &= \frac{2}{x} J_1(x) - J_0(x). \end{aligned}$$

Consequences and low-order checks II

Using the series for J_0 and J_1 ,

$$\begin{aligned}\frac{2}{x}J_1(x) - J_0(x) &= \left(1 - \frac{x^2}{8} + \frac{x^4}{192} + \cdots\right) - \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + \cdots\right) \\ &= \frac{x^2}{8} - \frac{x^4}{96} + \cdots \\ &= J_2(x),\end{aligned}$$

which checks the index and sign conventions.

The one-sided derivative recurrences can also be written in product form. From (6.2.5),

$$\begin{aligned}\frac{d}{dx} [x^m J_m(x)] &= mx^{m-1} J_m + x^m J'_m \\ &= mx^{m-1} J_m + x^m \left(J_{m-1} - \frac{m}{x} J_m \right) \\ &= x^m J_{m-1}(x),\end{aligned}$$

so

$$\frac{d}{dx} [x^m J_m(x)] = x^m J_{m-1}(x), \quad x \neq 0. \quad (6.2.9)$$

Consequences and low-order checks III

For the second product identity, use (6.2.6) explicitly:

$$\begin{aligned}\frac{d}{dx} [x^{-m} J_m(x)] &= -mx^{-m-1} J_m + x^{-m} J'_m \\ &= -mx^{-m-1} J_m + x^{-m} \left(\frac{m}{x} J_m - J_{m+1} \right) \\ &= -mx^{-m-1} J_m + mx^{-m-1} J_m - x^{-m} J_{m+1} \\ &= -x^{-m} J_{m+1}(x).\end{aligned}$$

Therefore

$$\frac{d}{dx} [x^{-m} J_m(x)] = -x^{-m} J_{m+1}(x), \quad x \neq 0. \quad (6.2.10)$$

Table of contents

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation**
- 4 Asymptotics
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series
- 7 Summary

Bessel's real integral representation I

Goal. Convert the complex full-period formula (6.1.9) into a real cosine integral on $[0, \pi]$.

Start from (renaming n as $m \in \mathbb{Z}$)

$$J_m(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - m\theta)} d\theta.$$

Split $\int_0^{2\pi} = \int_0^\pi + \int_\pi^{2\pi}$. In the second integral substitute $\theta = 2\pi - u$, so $d\theta = -du$ and the limits $\theta : \pi \rightarrow 2\pi$ become $u : \pi \rightarrow 0$; the minus sign flips them back to $u : 0 \rightarrow \pi$. Use

$$\sin(2\pi - u) = -\sin u, \quad -m(2\pi - u) = -2\pi m + mu.$$

The exponent becomes

$$\begin{aligned} i[x \sin(2\pi - u) - m(2\pi - u)] &= i[x(-\sin u) - 2\pi m + mu] \\ &= -i(x \sin u - mu) - i2\pi m. \end{aligned}$$

Bessel's real integral representation II

Hence

$$\begin{aligned} & \int_{\pi}^{2\pi} e^{i(x \sin \theta - m\theta)} d\theta \\ &= \int_0^{\pi} e^{i[x \sin(2\pi - u) - m(2\pi - u)]} du \\ &= \int_0^{\pi} e^{-i(x \sin u - mu)} e^{-i2\pi m} du \\ &= \int_0^{\pi} e^{-i(x \sin u - mu)} du, \end{aligned}$$

where $e^{-i2\pi m} = 1$ for integer m . Adding the first half-period integral $\int_0^{\pi} e^{i(x \sin u - mu)} du$ gives

$$\begin{aligned} \int_0^{\pi} [e^{i(x \sin u - mu)} + e^{-i(x \sin u - mu)}] du &= 2 \int_0^{\pi} \cos(x \sin u - mu) du \\ &= 2 \int_0^{\pi} \cos(mu - x \sin u) du, \end{aligned}$$

Bessel's real integral representation III

because cosine is even. Therefore

$$\begin{aligned} J_m(x) &= \frac{1}{2\pi} \cdot 2 \int_0^\pi \cos(mu - x \sin u) du \\ &= \frac{1}{\pi} \int_0^\pi \cos(mu - x \sin u) du. \end{aligned}$$

For $m \geq 0$ this is the classical real representation

$$J_m(x) = \frac{1}{\pi} \int_0^\pi \cos(m\theta - x \sin \theta) d\theta, \quad m \in \mathbb{Z}_{\geq 0}. \quad (6.3.1)$$

For $m = 0$ it reduces to

$$J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta) d\theta. \quad (6.3.2)$$

(The same formula for negative integer m follows from (6.1.7), or directly from (6.1.9).)

Contour form (Schl\"afli / Cauchy) I

The complex integral (6.1.9) is the unit-circle case of a more general contour formula. From the generating function (6.1.2) and Laurent extraction (6.1.3),

$$J_n(x) = \frac{1}{2\pi i} \oint_C e^{\frac{x}{2}(t-t^{-1})} t^{-n-1} dt, \quad n \in \mathbb{Z}, \quad (6.3.3)$$

where C is any positively oriented closed contour in $\mathbb{C} \setminus \{0\}$ with winding number one about the origin. (Uniform convergence of the Laurent series on the compact contour C permits termwise integration; only the residue at $t = 0$ of the t^{-1} term survives.)

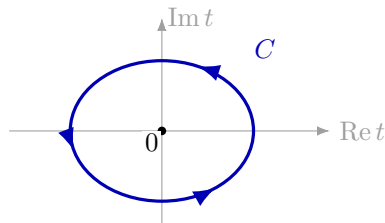
Parameterizing C by $t = e^{i\theta}$ recovers (6.1.9) immediately:

$$dt = it d\theta, \quad \frac{x}{2}(t - t^{-1}) = ix \sin \theta, \quad t^{-n-1} dt = ie^{-in\theta} d\theta.$$

For noninteger order the factor $t^{-\nu-1}$ is multivalued and a branch-compatible (nonclosed) contour is required; we use only the integer-order closed form above.

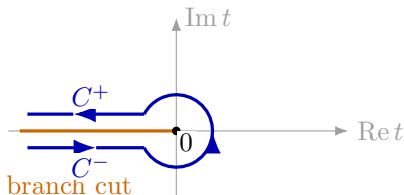
Contours in the Schlöfli representation

integer order



t^{-n-1} single-valued
winding number $+1$

noninteger order



$t^{-\nu-1} = e^{-(\nu+1) \text{Log } t}$
branch must be fixed

For integer n , t^{-n-1} is single-valued and Cauchy's coefficient formula only requires a closed contour with winding number one around 0. For noninteger order, a branch of $\text{Log } t$ and a contour compatible with its branch cut are required; C^+ and C^- label the upper and lower sides of that cut, respectively.

Table of contents

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation
- 4 Asymptotics**
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series
- 7 Summary

Small x : leading term and first corrections I

Step 1: factor out the leading term. For fixed $m \in \mathbb{Z}_{\geq 0}$, factor the $k = 0$ term out of the power series (6.1.4). The first three terms are

$$J_m(x) = \frac{1}{m!} \left(\frac{x}{2}\right)^m - \frac{1}{(m+1)!} \left(\frac{x}{2}\right)^{m+2} + \frac{1}{2!(m+2)!} \left(\frac{x}{2}\right)^{m+4} + O(x^{m+6}).$$

Divide the second term by the first:

$$\begin{aligned} \frac{-\frac{1}{(m+1)!} (x/2)^{m+2}}{\frac{1}{m!} (x/2)^m} &= -\frac{m!}{(m+1)!} \left(\frac{x}{2}\right)^2 \\ &= -\frac{x^2}{4(m+1)}. \end{aligned}$$

Small x : leading term and first corrections II

Divide the third term by the first:

$$\frac{\frac{1}{2!(m+2)!}(x/2)^{m+4}}{\frac{1}{m!}(x/2)^m} = \frac{m!}{2(m+2)!} \left(\frac{x}{2}\right)^4 \\ = \frac{x^4}{32(m+1)(m+2)}.$$

Hence

$$J_m(x) = \frac{1}{m!} \left(\frac{x}{2}\right)^m \left[1 - \frac{x^2}{4(m+1)} + \frac{x^4}{32(m+1)(m+2)} + O(x^6) \right]. \quad (6.4.1)$$

In particular, writing $m! = \Gamma(m+1)$ (which anticipates the noninteger-order case),

$$J_m(x) = \frac{1}{\Gamma(m+1)} \left(\frac{x}{2}\right)^m + O(x^{m+2}), \quad x \rightarrow 0, \quad (6.4.2)$$

with m fixed. This gives $J_0(0) = 1$ and $J_m(0) = 0$ for $m \geq 1$.

Large x : stationary phase on the unit circle I

For large x with fixed order, the unit-circle integral (6.1.9) is an oscillatory integral. Chapter 2 developed steepest descent for $\int g(z)e^{sf(z)} dz$ and recorded its real oscillatory case *stationary phase*: the leading contribution as $x \rightarrow +\infty$ comes from neighborhoods of points where the phase is stationary; away from those points, repeated integration by parts produces an arbitrarily small power of x^{-1} . Here we *apply* that tool to Bessel's integral—we do not re-derive the master Gaussian formula.

Step 1: locate the stationary points. For fixed integer n , write

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} a(\theta) e^{ix\phi(\theta)} d\theta, \quad a(\theta) = e^{-in\theta}, \quad \phi(\theta) = \sin \theta.$$

The stationary points of the large phase $x\phi(\theta)$ satisfy

$$\phi'(\theta) = \cos \theta = 0,$$

so

$$\theta_+ = \frac{\pi}{2}, \quad \theta_- = \frac{3\pi}{2}.$$

They correspond to $t = e^{i\theta} = i$ and $t = -i$ on the unit circle. Choose a smooth 2π -periodic partition of unity

$$\eta_+(\theta) + \eta_-(\theta) + \eta_0(\theta) = 1,$$

Large x : stationary phase on the unit circle II

where η_+ and η_- are supported in small neighborhoods of θ_+ and θ_- , respectively. On the support of η_0 , there is a constant $c > 0$ such that $|\phi'(\theta)| \geq c$, so $\phi'(\theta) \neq 0$ there. Since $\frac{d}{d\theta} e^{ix\phi} = ix\phi'(\theta) e^{ix\phi}$, we may solve for the exponential,

$$e^{ix\phi(\theta)} = \frac{1}{ix\phi'(\theta)} \frac{d}{d\theta} e^{ix\phi(\theta)},$$

and one integration by parts gives

$$\int_0^{2\pi} \eta_0 a e^{ix\phi} d\theta = -\frac{1}{ix} \int_0^{2\pi} \frac{d}{d\theta} \left(\frac{\eta_0 a}{\phi'} \right) e^{ix\phi} d\theta.$$

There is no boundary term because every factor is 2π -periodic. Repeating this operation N times gives an $O(x^{-N})$ contribution for every fixed N . Hence only the two stationary neighborhoods contribute to the algebraic asymptotic expansion.

Step 2: analyze the first saddle. Near $\theta_+ = \pi/2$, set $\theta = \pi/2 + u$. Choose the local cutoff η_+ to be even in u and equal to 1 near $u = 0$. Then

$$\begin{aligned} \sin\left(\frac{\pi}{2} + u\right) &= \cos u = 1 - \frac{u^2}{2} + O(u^4), \\ e^{ix \sin(\pi/2+u)} &= e^{ix \cos u} = e^{ix} e^{-ixu^2/2 + O(xu^4)}, \\ e^{-in(\pi/2+u)} &= e^{-in\pi/2} e^{-inu} = e^{-in\pi/2} \left(1 - inu - \frac{n^2 u^2}{2} + O(u^3) \right). \end{aligned}$$

Large x : stationary phase on the unit circle III

Write $\tilde{\eta}_+(u) := \eta_+(\frac{\pi}{2} + u)$. By construction $\tilde{\eta}_+$ is even, equals 1 for $|u| \leq \delta/2$, and vanishes for $|u| \geq \delta$, where $\delta < \pi/2$ is the radius of the neighborhood supporting η_+ . The localized contribution is then the explicit integral

$$\begin{aligned} J_n^{(+)}(x) &:= \frac{1}{2\pi} \int_0^{2\pi} \eta_+(\theta) e^{-in\theta} e^{ix \sin \theta} d\theta \\ &= \frac{e^{ix} e^{-in\pi/2}}{2\pi} \int_{-\infty}^{\infty} \tilde{\eta}_+(u) e^{-inu} e^{ix(\cos u - 1)} du, \end{aligned}$$

where the second line is the substitution $\theta = \frac{\pi}{2} + u$ ($d\theta = du$) combined with the two factorizations $e^{-in\theta} = e^{-in\pi/2} e^{-inu}$ and $e^{ix \sin \theta} = e^{ix \cos u} = e^{ix} e^{ix(\cos u - 1)}$ displayed above, and the range may be written as all of $\mathbb{R}$ because $\tilde{\eta}_+$ vanishes for $|u| \geq \delta$. Scale $u = v/\sqrt{x}$, so $du = dv/\sqrt{x}$. Then

$$\begin{aligned} x \left[\sin \left(\frac{\pi}{2} + u \right) - 1 \right] &= -\frac{v^2}{2} + O\left(\frac{v^4}{x}\right), \\ e^{-inu} &= 1 - \frac{inv}{\sqrt{x}} + O\left(\frac{v^2}{x}\right), \\ e^{ix \sin(\pi/2+u)} &= e^{ix} e^{-iv^2/2 + O(v^4/x)}. \end{aligned}$$

Large x : stationary phase on the unit circle IV

The term proportional to $v/\sqrt{x}$ is odd and integrates to zero against the even local cutoff and the even quadratic phase. All first nonzero corrections are therefore of relative order x^{-1} ; after the Jacobian factor $x^{-1/2}$, their absolute order is $x^{-3/2}$.

With these expansions the localized integral has become

$$J_n^{(+)}(x) = \frac{e^{ix} e^{-in\pi/2}}{2\pi} \int_{-\infty}^{\infty} \tilde{\eta}_+(u) e^{-ixu^2/2} du + O(x^{-3/2}),$$

and it remains to delete the cutoff. On the support of $1 - \tilde{\eta}_+$ we have $|u| \geq \delta/2$, so there the quadratic phase is nonstationary and the device of Step 1 applies once more: from $\frac{d}{du} e^{-ixu^2/2} = -ixu e^{-ixu^2/2}$ and one integration by parts,

$$\int_{-\infty}^{\infty} (1 - \tilde{\eta}_+(u)) e^{-ixu^2/2} du = \frac{1}{ix} \int_{-\infty}^{\infty} \frac{d}{du} \left(\frac{1 - \tilde{\eta}_+(u)}{u} \right) e^{-ixu^2/2} du,$$

with no boundary term because $(1 - \tilde{\eta}_+(u))/u \rightarrow 0$ as $|u| \rightarrow \infty$. The new amplitude is smooth (it vanishes identically near $u = 0$, so the division by u is harmless) and its derivative is $O(u^{-2})$ at infinity, hence absolutely integrable; the step may therefore be repeated, each repetition producing another factor x^{-1} . The difference between the cutoff integral and the full one is thus $O(x^{-N})$ for every fixed N , which is beyond every term we retain. (As in the damped-limit convention introduced next, whole-line

Large x : stationary phase on the unit circle V

oscillatory integrals are understood as $\varepsilon \downarrow 0$ limits.) Thus the leading local contribution is

$$\begin{aligned} J_n^{(+)}(x) &= \frac{e^{ix} e^{-in\pi/2}}{2\pi} \int_{-\infty}^{\infty} e^{-ixu^2/2} du + O(x^{-3/2}) \\ &= \frac{e^{ix} e^{-in\pi/2}}{2\pi\sqrt{x}} \int_{-\infty}^{\infty} e^{-iv^2/2} dv + O(x^{-3/2}). \end{aligned}$$

To define the oscillatory Gaussian unambiguously, introduce a damping factor $e^{-\varepsilon u^2}$ and let $\varepsilon \downarrow 0$. First record the Gaussian integral with a complex parameter. For real $a > 0$, the substitution $u = t/\sqrt{a}$ in Chapter 3's Gaussian integral $\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$ gives

$$\int_{-\infty}^{\infty} e^{-au^2} du = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\frac{\pi}{a}}.$$

This persists for complex a in the right half-plane. Indeed $|e^{-au^2}| = e^{-(\operatorname{Re} a)u^2}$, so on $\operatorname{Re} a > 0$ the integral converges absolutely and locally uniformly and therefore defines an analytic function of a there; $\sqrt{\pi/a}$, taken on the principal branch (positive for $a > 0$), is analytic there as well; and the two agree on the positive real axis. By the

Large x : stationary phase on the unit circle VI

identity theorem (Chapter 2, used the same way in Chapter 3) they agree on the whole half-plane:

$$\int_{-\infty}^{\infty} e^{-au^2} du = \sqrt{\frac{\pi}{a}}, \quad \operatorname{Re} a > 0, \quad (6.4.3)$$

with the principal square root. For $\varepsilon > 0$ the parameter $a = \varepsilon + ix/2$ has $\operatorname{Re} a = \varepsilon > 0$, so (6.4.3) applies:

$$\begin{aligned} I_{\varepsilon}^{-}(x) &:= \int_{-\infty}^{\infty} e^{-(\varepsilon+ix/2)u^2} du \\ &= \sqrt{\frac{\pi}{\varepsilon + ix/2}}. \end{aligned}$$

On the principal square-root branch, as $\varepsilon \downarrow 0$ with $x > 0$ fixed,

$$\begin{aligned} \varepsilon + \frac{ix}{2} &= \frac{x}{2} e^{i\pi/2} (1 + o(1)), \\ \left(\varepsilon + \frac{ix}{2}\right)^{-1/2} &\rightarrow \sqrt{\frac{2}{x}} e^{-i\pi/4}. \end{aligned}$$

Large x : stationary phase on the unit circle VII

Hence

$$I_{\epsilon}^{-}(x) \longrightarrow \sqrt{\pi} \cdot \sqrt{\frac{2}{x}} e^{-i\pi/4} = e^{-i\pi/4} \sqrt{\frac{2\pi}{x}},$$

and the Fresnel integral, understood as this damped limit, is

$$\int_{-\infty}^{\infty} e^{-ixu^2/2} du = e^{-i\pi/4} \sqrt{\frac{2\pi}{x}}, \quad x > 0. \quad (6.4.4)$$

Therefore

$$\begin{aligned} J_n^{(+)}(x) &= \frac{e^{ix} e^{-in\pi/2}}{2\pi} \cdot e^{-i\pi/4} \sqrt{\frac{2\pi}{x}} + O(x^{-3/2}) \\ &= \frac{1}{\sqrt{2\pi x}} e^{i(x-n\pi/2-\pi/4)} + O(x^{-3/2}), \end{aligned}$$

where the last line uses $\sqrt{2\pi/x}/(2\pi) = 1/\sqrt{2\pi x}$.

Large x : stationary phase on the unit circle VIII

Step 3: analyze the second saddle. Near $\theta_- = 3\pi/2$, set $\theta = 3\pi/2 + u$ and choose the corresponding local cutoff even in u . Then

$$\begin{aligned}\sin\left(\frac{3\pi}{2} + u\right) &= -\cos u = -1 + \frac{u^2}{2} + O(u^4), \\ e^{ix \sin(3\pi/2+u)} &= e^{-ix} e^{ixu^2/2 + O(xu^4)}, \\ e^{-in(3\pi/2+u)} &= e^{-i3n\pi/2} e^{-inu}.\end{aligned}$$

For integer n ,

$$e^{-i3n\pi/2} = e^{in\pi/2} e^{-i2\pi n} = e^{in\pi/2}.$$

The same scaling $u = v/\sqrt{x}$ shows that the odd amplitude correction integrates to zero and that the first nonzero omitted terms are $O(x^{-3/2})$. Thus

$$J_n^{(-)}(x) = \frac{e^{-ix} e^{in\pi/2}}{2\pi} \int_{-\infty}^{\infty} e^{ixu^2/2} du + O(x^{-3/2}).$$

The conjugate (damped) calculation of (6.4.4) gives

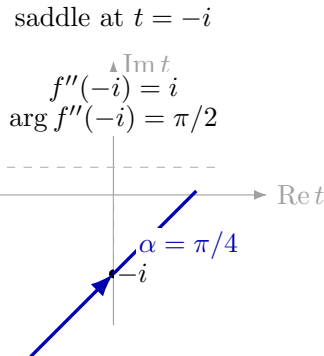
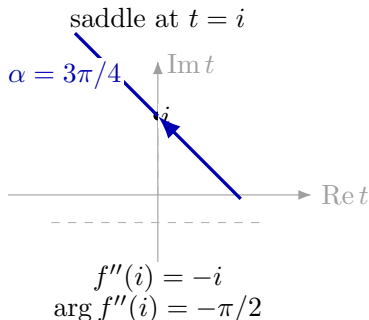
$$\int_{-\infty}^{\infty} e^{ixu^2/2} du = e^{i\pi/4} \sqrt{\frac{2\pi}{x}}, \quad x > 0. \quad (6.4.5)$$

Large x : stationary phase on the unit circle IX

Hence

$$J_n^{(-)}(x) = \frac{1}{\sqrt{2\pi x}} e^{-i(x - n\pi/2 - \pi/4)} + O(x^{-3/2}).$$

Steepest descent and the unit-circle saddles



In the complex t -plane the same two stationary points are the saddles $t = \pm i$ of $f(t) = \frac{1}{2}(t - t^{-1})$ in the Schlöfli integrand $e^{xf(t)}t^{-n-1}$. Here

$$f'(t) = \frac{1}{2} \left(1 + \frac{1}{t^2} \right), \quad f''(t) = -\frac{1}{t^3},$$

so $f'(t) = 0$ exactly when $t^2 = -1$, i.e. $t = \pm i$, which are the same two points found on the unit circle. At those points

Large x : combining the two saddle contributions I

Define the phase

$$\chi_n(x) = x - \frac{n\pi}{2} - \frac{\pi}{4}.$$

By the partition of unity $\eta_+ + \eta_- + \eta_0 = 1$ the exact splitting of Bessel's integral is $J_n = J_n^{(+)} + J_n^{(-)} + J_n^{(0)}$, where

$$J_n^{(0)}(x) = \frac{1}{2\pi} \int_0^{2\pi} \eta_0(\theta) e^{-in\theta} e^{ix \sin \theta} d\theta$$

is the nonstationary piece estimated in Step 1. Adding the two stationary-phase contributions therefore gives

$$\begin{aligned} J_n(x) &= J_n^{(+)}(x) + J_n^{(-)}(x) + J_n^{(0)}(x) \\ &= J_n^{(+)}(x) + J_n^{(-)}(x) + O(x^{-2}) \\ &= \frac{1}{\sqrt{2\pi x}} (e^{i\chi_n(x)} + e^{-i\chi_n(x)}) + O(x^{-3/2}), \end{aligned}$$

where the second line is Step 1 with $N = 2$ (any $N \geq 2$ would do, since the remainder we are about to write is only $O(x^{-3/2})$). Using

$$e^{i\chi} + e^{-i\chi} = 2 \cos \chi,$$

Large x : combining the two saddle contributions II

we obtain

$$\begin{aligned} J_n(x) &= \frac{2}{\sqrt{2\pi x}} \cos \chi_n(x) + O(x^{-3/2}) \\ &= \sqrt{\frac{2}{\pi x}} \cos \left(x - \frac{n\pi}{2} - \frac{\pi}{4} \right) + O(x^{-3/2}). \end{aligned}$$

Thus the leading large-argument expansion is

$$J_n(x) = \sqrt{\frac{2}{\pi x}} \cos \left(x - \frac{n\pi}{2} - \frac{\pi}{4} \right) + O(x^{-3/2}), \quad x \rightarrow +\infty, \quad (6.4.6)$$

for fixed integer n . The equality with an absolute error term is preferable to writing a ratio asymptotic $J_n \sim \sqrt{2/(\pi x)} \cos \chi_n$, because the leading cosine has infinitely many zeros.

For non-integer fixed order ν the same leading cosine law

$$J_\nu(x) = \sqrt{\frac{2}{\pi x}} \cos \left(x - \frac{\nu\pi}{2} - \frac{\pi}{4} \right) + O(x^{-3/2}), \quad x \rightarrow +\infty,$$

holds, with the remainder uniform for ν in any bounded subset of the complex plane.

The derivation just given does *not* cover it: the integration by parts of Step 1 discarded its boundary terms because $e^{-in\theta}$ is 2π -periodic, which fails as soon as $\nu \notin \mathbb{Z}$. The proper route is steepest descent applied to the Hankel/Schlöfli-type

Large x : combining the two saddle contributions III

contour of the previous section, on which the multivalued factor $t^{-\nu-1}$ is admissible; that computation meets the same two saddles $t = \pm i$ and returns the same phase and amplitude, and it is also the source of the uniformity in ν . We quote it from the standard references on Bessel asymptotics (Watson's treatise; DLMF Chapter 10) and do not repeat the contour derivation here.

One caution, since the temptation is natural: this cannot be obtained by *continuing the integer-order result in ν* . Both $J_\nu(x)$ and $\sqrt{2/(\pi x)} \cos \chi_\nu(x)$ are entire in ν , so their difference is entire in ν for each fixed x ; but a size estimate on the discrete set $\nu \in \mathbb{Z}$ says nothing in between. For instance $E(\nu, x) = \sin(\pi\nu)\sqrt{x}$ is entire in ν and vanishes identically at every integer order, yet $E(\frac{1}{2}, x) = \sqrt{x}$ grows without bound. Analytic continuation transports identities, not bounds sampled along a discrete set; the uniform bound has to come from the contour representation itself.

We will use the displayed formula, and the uniformity just stated, when deriving the large- x expansion of Y_ν below.

Table of contents

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation
- 4 Asymptotics
- 5 Neumann functions**
- 6 Hankel functions and Fourier–Bessel series
- 7 Summary

Reduction of order: constructing a second solution I

Step 1: reduce order. Divide Bessel's equation by x^2 on an interval with $x > 0$:

$$y'' + \frac{1}{x}y' + \left(1 - \frac{m^2}{x^2}\right)y = 0. \quad (6.5.1)$$

Let J_m be one solution and seek a second solution in the form

$$y_2(x) = v(x)J_m(x).$$

Differentiate:

$$\begin{aligned} y_2' &= v'J_m + vJ_m', \\ y_2'' &= v''J_m + 2v'J_m' + vJ_m''. \end{aligned}$$

Substitute into (6.5.1):

$$\begin{aligned} 0 &= v''J_m + 2v'J_m' + vJ_m'' + \frac{1}{x}(v'J_m + vJ_m') + \left(1 - \frac{m^2}{x^2}\right)vJ_m \\ &= v''J_m + v' \left(2J_m' + \frac{1}{x}J_m\right) \\ &\quad + v \left[J_m'' + \frac{1}{x}J_m' + \left(1 - \frac{m^2}{x^2}\right)J_m \right]. \end{aligned}$$

The bracket multiplying v is zero because J_m solves Bessel's equation. Thus

$$v''J_m + v' \left(2J_m' + \frac{1}{x}J_m\right) = 0.$$

Reduction of order: constructing a second solution II

Step 2: solve for the new factor. Set $w = v'$. On an interval where $J_m \neq 0$ and $w \neq 0$,

$$\begin{aligned}w' J_m + w \left(2J'_m + \frac{1}{x} J_m \right) &= 0, \\ \frac{w'}{w} &= -2 \frac{J'_m}{J_m} - \frac{1}{x}.\end{aligned}$$

Integrating gives

$$\begin{aligned}\log |w| &= -2 \log |J_m| - \log x + C, \\ w = v' &= \frac{C_1}{x J_m^2(x)}.\end{aligned}$$

Taking $C_1 = 1$ and integrating once more,

$$\begin{aligned}v(x) &= C_2 + \int_{x_*}^x \frac{ds}{s J_m^2(s)}, \\ y_2(x) &= C_2 J_m(x) + J_m(x) \int_{x_*}^x \frac{ds}{s J_m^2(s)}.\end{aligned}$$

Hence the general solution on a zero-free interval is

$$y(x) = A J_m(x) + B J_m(x) \int_{x_*}^x \frac{ds}{s J_m^2(s)}. \quad (6.5.2)$$

Reduction of order: constructing a second solution III

For functions u and v , use the Wronskian convention

$$W[u, v] = uv' - u'v.$$

The constructed second solution then has

$$\begin{aligned} W[J_m, y_2] &= J_m y_2' - J_m' y_2 \\ &= J_m(v' J_m + v J_m') - J_m'(v J_m) \\ &= v' J_m^2 \\ &= \frac{1}{x}. \end{aligned}$$

Step 3: normalize the second solution. The conventional Neumann function is normalized by

$$W[J_m, Y_m] = \frac{2}{\pi x}. \quad (6.5.3)$$

Here Y_m is the Neumann function of integer order m , built on the next two frames as the limit $\lim_{\nu \rightarrow m} Y_\nu$ of the noninteger-order function (6.5.4). That this limit again solves (6.5.1), and that it satisfies (6.5.3), are proved there; we quote the two facts now only because they are what fixes the normalization of the second solution. The constructed solution $y_2(x) = J_m(x) \int_{x_*}^x ds/(sJ_m^2(s))$ has $W[J_m, y_2] = 1/x$ by the

Reduction of order: constructing a second solution IV

computation above, while (6.5.3) gives $W[J_m, \frac{\pi}{2} Y_m] = \frac{\pi}{2} \cdot \frac{2}{\pi x} = \frac{1}{x}$. Subtracting, $W[J_m, y_2 - \frac{\pi}{2} Y_m] = 0$; two solutions of the same second-order linear equation whose Wronskian vanishes are linearly dependent, so $y_2 - \frac{\pi}{2} Y_m = c J_m$ for some constant c . Therefore the reduction-of-order solution equals $(\pi/2) Y_m$ plus a constant multiple of J_m . The apparent singularities of the integral at zeros of J_m only limit this local formula; the resulting second solution can be continued across such points.

Noninteger order: definition and independence of Y_ν I

Bessel's equation depends on the order only through ν^2 . Therefore, if $J_\nu(x)$ is a solution, then $J_{-\nu}(x)$ is also a solution. For fixed real $\nu \notin \mathbb{Z}$, define

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)}. \quad (6.5.4)$$

To verify independence, first compute $W[J_\nu, J_{-\nu}]$ from their leading small- x terms. For $\nu \notin \mathbb{Z}$, retaining only the $k = 0$ term of the series (6.1.5) gives $(x/2)^{\pm\nu}/\Gamma(1 \pm \nu)$, so that

$$\begin{aligned} J_\nu(x) &\sim Ax^\nu, & A &= \frac{2^{-\nu}}{\Gamma(\nu+1)}, \\ J_{-\nu}(x) &\sim Bx^{-\nu}, & B &= \frac{2^\nu}{\Gamma(1-\nu)}. \end{aligned}$$

Thus

$$\begin{aligned} W[J_\nu, J_{-\nu}] &= J_\nu J'_{-\nu} - J'_\nu J_{-\nu} \\ &\sim (Ax^\nu)(-\nu Bx^{-\nu-1}) - (\nu Ax^{\nu-1})(Bx^{-\nu}) \\ &= -2\nu AB x^{-1}. \end{aligned}$$

Noninteger order: definition and independence of Y_ν II

This leading calculation determines the exact Wronskian constant. With the Wronskian convention introduced above, take two solutions of

$$y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right)y = 0,$$

differentiate the Wronskian:

$$\begin{aligned} W' &= \frac{d}{dx}(y_1 y_2' - y_1' y_2) \\ &= y_1' y_2' + y_1 y_2'' - y_1'' y_2 - y_1' y_2' \\ &= y_1 y_2'' - y_1'' y_2. \end{aligned}$$

Using

$$y_j'' = -\frac{1}{x}y_j' - \left(1 - \frac{\nu^2}{x^2}\right)y_j, \quad j = 1, 2,$$

Noninteger order: definition and independence of Y_ν III

we obtain

$$\begin{aligned} W' &= y_1 \left[-\frac{1}{x} y_2' - \left(1 - \frac{\nu^2}{x^2} \right) y_2 \right] - \left[-\frac{1}{x} y_1' - \left(1 - \frac{\nu^2}{x^2} \right) y_1 \right] y_2 \\ &= -\frac{1}{x} (y_1 y_2' - y_1' y_2) \\ &= -\frac{1}{x} W. \end{aligned}$$

Therefore

$$\begin{aligned} \frac{d}{dx} [xW(x)] &= W(x) + xW'(x) = W(x) - W(x) = 0, \\ xW(x) &= C_\nu, \\ W(x) &= \frac{C_\nu}{x}. \end{aligned}$$

Noninteger order: definition and independence of Y_ν IV

The small- x limit above therefore fixes $C_\nu = -2\nu AB$. Now

$$\begin{aligned} AB &= \frac{1}{\Gamma(\nu+1)\Gamma(1-\nu)} \\ &= \frac{1}{\nu\Gamma(\nu)\Gamma(1-\nu)} \\ &= \frac{\sin(\pi\nu)}{\pi\nu}, \end{aligned}$$

where the last step uses Euler's reflection formula $\Gamma(\nu)\Gamma(1-\nu) = \pi/\sin(\pi\nu)$. Hence

$$W[J_\nu, J_{-\nu}] = -\frac{2\sin(\pi\nu)}{\pi x}. \quad (6.5.5)$$

Noninteger order: definition and independence of Y_ν

Using linearity of the Wronskian in its second argument together with $W[J_\nu, J_\nu] = 0$, the Wronskian of (6.5.4) is therefore

$$\begin{aligned} W[J_\nu, Y_\nu] &= \frac{1}{\sin(\pi\nu)} W[J_\nu, J_\nu \cos(\pi\nu) - J_{-\nu}] \\ &= \frac{1}{\sin(\pi\nu)} [\cos(\pi\nu) W[J_\nu, J_\nu] - W[J_\nu, J_{-\nu}]] \\ &= -\frac{1}{\sin(\pi\nu)} W[J_\nu, J_{-\nu}] \\ &= \frac{2}{\pi x}, \end{aligned}$$

which establishes independence and the normalization $W[J_\nu, Y_\nu] = 2/(\pi x)$ for every noninteger ν .

The integer case. Equation (6.5.3) was stated at integer order, and the computation just made cannot be run there, since it divides by $\sin(\pi\nu)$. We reach it by letting $\nu \rightarrow m$. Fix $x > 0$. Each term of (6.1.5) is entire in ν , because $1/\Gamma(k + \nu + 1)$ is, and the ratio bound on that frame, $|T_{k+1}/T_k| = |x|^2/(4(k+1)|k + \nu + 1|)$, is uniform for ν in a compact set as well as for x in a compact set. Hence $\nu \mapsto J_{\pm\nu}(x)$ and $\nu \mapsto \partial_x J_{\pm\nu}(x)$ are analytic in ν , and so are

$$N(\nu, x) = J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x), \quad \partial_x N(\nu, x), \quad S(\nu) = \sin(\pi\nu).$$

Noninteger order: definition and independence of Y_ν VI

At $\nu = m$ the relations $J_{-m} = (-1)^m J_m$ and $\cos(\pi m) = (-1)^m$ give $N(m, x) = 0$ for every x , hence also $\partial_x N(m, x) = 0$, while $S(m) = 0$ with $S'(m) = \pi(-1)^m \neq 0$. Both quotients

$$Y_\nu(x) = \frac{N(\nu, x)}{S(\nu)}, \quad \partial_x Y_\nu(x) = \frac{\partial_x N(\nu, x)}{S(\nu)}$$

therefore have removable singularities at $\nu = m$: each extends continuously in ν , the value of the first being the l'Hôpital limit (6.5.6) computed on the next frame.

Letting $\nu \rightarrow m$ in the identity just proved,

$$W[J_m, Y_m] = \lim_{\nu \rightarrow m} (J_\nu \partial_x Y_\nu - \partial_x J_\nu Y_\nu) = \frac{2}{\pi x}, \quad m \in \mathbb{Z},$$

which is (6.5.3) as used in Step 3. In particular the Wronskian is nonzero, so J_m and Y_m are independent at integer order too.

Integer order: the l'Hôpital limit without ambiguity I

At an integer m , both numerator and denominator of (6.5.4) vanish because

$$J_{-m} = (-1)^m J_m, \quad \cos(\pi m) = (-1)^m, \quad \sin(\pi m) = 0.$$

Introduce the order-derivative notation

$$D_\mu J_\mu(x) := \frac{\partial}{\partial \mu} J_\mu(x).$$

For fixed $x > 0$ and a fixed logarithm branch, the series for $J_\nu(x)$ is analytic in ν , so the order derivatives below exist. Let

$$N(\nu) = J_\nu(x) \cos(\pi \nu) - J_{-\nu}(x), \quad S(\nu) = \sin(\pi \nu).$$

Differentiate the numerator. The first product gives

$$\frac{d}{d\nu} [J_\nu \cos(\pi \nu)] = D_\nu J_\nu \cos(\pi \nu) - \pi J_\nu \sin(\pi \nu).$$

For the second term, the chain rule gives

$$\begin{aligned} \frac{d}{d\nu} J_{-\nu}(x) &= D_\mu J_\mu(x)|_{\mu=-\nu} \frac{d(-\nu)}{d\nu} \\ &= - D_\mu J_\mu(x)|_{\mu=-\nu}. \end{aligned}$$

Therefore

$$N'(\nu) = D_\nu J_\nu \cos(\pi \nu) - \pi J_\nu \sin(\pi \nu) + D_\mu J_\mu|_{\mu=-\nu}.$$

Integer order: the l'Hôpital limit without ambiguity II

At $\nu = m$,

$$N'(m) = (-1)^m D_\nu J_\nu(x)|_{\nu=m} + D_\nu J_\nu(x)|_{\nu=-m},$$

$$S'(m) = \pi \cos(\pi m) = \pi(-1)^m \neq 0.$$

L'Hôpital's rule now gives

$$\begin{aligned} Y_m(x) &= \frac{N'(m)}{S'(m)} \\ &= \frac{(-1)^m D_\nu J_\nu|_{\nu=m} + D_\nu J_\nu|_{\nu=-m}}{\pi(-1)^m} \\ &= \frac{1}{\pi} \left[D_\nu J_\nu(x)|_{\nu=m} + (-1)^m D_\nu J_\nu(x)|_{\nu=-m} \right]. \end{aligned}$$

Thus

$$Y_m(x) = \frac{1}{\pi} \left[\left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=m} + (-1)^m \left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=-m} \right]. \quad (6.5.6)$$

This formula explicitly distinguishes an order derivative from the total derivative of the composite function $J_{-\nu}$; omitting that chain-rule sign changes the result.

Integer order: the l'Hôpital limit without ambiguity III

The limit is still a solution. Step 3 of the reduction-of-order frame treated Y_m as a solution of Bessel's equation, so we check that. Write

$$\mathcal{L}_\nu[y] = y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right)y,$$

so that Bessel's equation of order ν is $\mathcal{L}_\nu[y] = 0$, and $\mathcal{L}_\nu[J_\nu] = 0$ for every ν and every $x > 0$. The check is needed because a single order derivative is *not* a solution: differentiate the identity $\mathcal{L}_\nu[J_\nu] = 0$ with respect to ν , interchanging ∂_ν with the x -derivatives (legitimate because the series and its x -derivatives converge locally uniformly in ν). The explicit ν -dependence of $\mathcal{L}_\nu$ contributes $-2\nu J_\nu/x^2$, so

$$0 = \partial_\nu(\mathcal{L}_\nu[J_\nu]) = \mathcal{L}_\nu[D_\nu J_\nu] - \frac{2\nu}{x^2}J_\nu, \quad \text{that is} \quad \mathcal{L}_\nu[D_\nu J_\nu] = \frac{2\nu}{x^2}J_\nu.$$

Evaluate this at $\nu = m$ and at $\nu = -m$, using $\mathcal{L}_{-m} = \mathcal{L}_m$ (the operator depends on the order only through ν^2) and $J_{-m} = (-1)^m J_m$:

$$\begin{aligned}\mathcal{L}_m[D_\nu J_\nu|_{\nu=m}] &= \frac{2m}{x^2}J_m, \\ \mathcal{L}_m[D_\nu J_\nu|_{\nu=-m}] &= -\frac{2m}{x^2}J_{-m} = -(-1)^m \frac{2m}{x^2}J_m.\end{aligned}$$

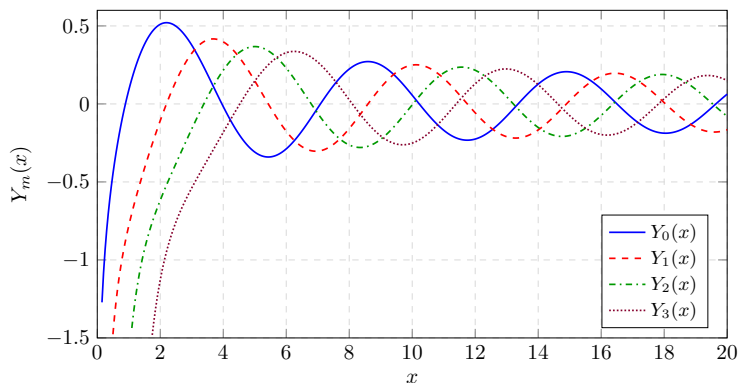
Integer order: the l'Hôpital limit without ambiguity IV

By linearity of $\mathcal{L}_m$, (6.5.6) therefore gives

$$\pi \mathcal{L}_m[Y_m] = \frac{2m}{x^2} J_m + (-1)^m \left(-(-1)^m \frac{2m}{x^2} J_m \right) = \frac{2m}{x^2} J_m - \frac{2m}{x^2} J_m = 0.$$

The inhomogeneities cancel exactly, so Y_m solves Bessel's equation of order m , as Step 3 assumed.

Bessel functions of the second kind



The Neumann functions are independent of J_m and are singular at $x = 0$. Away from the origin, J_m and Y_m have the same $x^{-1/2}$ envelope and phases shifted by $\pi/2$.

Small- x behavior of Y_m

First let $\nu > 0$ be fixed and noninteger. From the first term of the series,

$$J_\nu(x) = \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu [1 + O(x^2)],$$
$$J_{-\nu}(x) = \frac{1}{\Gamma(1-\nu)} \left(\frac{x}{2}\right)^{-\nu} [1 + O(x^2)].$$

Their magnitudes have orders x^ν and $x^{-\nu}$, respectively. Since $x^\nu/x^{-\nu} = x^{2\nu} \rightarrow 0$, the $J_{-\nu}$ term dominates in (6.5.4). Therefore

$$Y_\nu(x) = -\frac{1}{\Gamma(1-\nu)\sin(\pi\nu)} \left(\frac{x}{2}\right)^{-\nu} [1 + o(1)].$$

Euler's reflection formula gives

$$\Gamma(\nu)\Gamma(1-\nu) = \frac{\pi}{\sin(\pi\nu)},$$
$$\Gamma(1-\nu)\sin(\pi\nu) = \frac{\pi}{\Gamma(\nu)},$$
$$\frac{1}{\Gamma(1-\nu)\sin(\pi\nu)} = \frac{\Gamma(\nu)}{\pi}.$$

Consequently,

$$Y_\nu(x) = -\frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x}\right)^\nu [1 + o(1)], \quad x \rightarrow 0^+, \quad (6.5.7)$$

Small- x behavior of Y_m II

for fixed positive noninteger ν .

We now derive the positive-integer result directly, rather than interchanging the limits $x \rightarrow 0$ and $\nu \rightarrow m$. Let $m \geq 1$. Differentiating the series termwise gives $\partial_\nu J_\nu = \sum_{k \geq 0} \frac{(-1)^k}{k! \Gamma(k+\nu+1)} (x/2)^{2k+\nu} [L - \psi(k+\nu+1)]$, where $L = \log(x/2)$ and $\psi = \Gamma'/\Gamma$ is the digamma function. At $\nu = m$ the smallest power is the $k = 0$ term $\frac{1}{\Gamma(m+1)} (x/2)^m [L - \psi(m+1)]$; since L grows like $\log x$ and every $k \geq 1$ term carries an extra $(x/2)^{2k}$, the order-derivative formula (6.5.6) has

$$\left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=m} = O(x^m |\log x|).$$

The dominant term comes from the $k = 0$ term of the derivative at $\nu = -m$. To evaluate it, write

$$\frac{1}{\Gamma(\nu+1)} = \frac{(\nu+1)(\nu+2)\cdots(\nu+m)}{\Gamma(\nu+m+1)}.$$

At $\nu = -m$ exactly one factor of the numerator, namely $\nu + m$, vanishes. Write the right-hand side as $P(\nu)h(\nu)$ with

$$P(\nu) = (\nu+1)(\nu+2)\cdots(\nu+m), \quad h(\nu) = \frac{1}{\Gamma(\nu+m+1)},$$

Small- x behavior of Y_m III

where h is differentiable near $\nu = -m$ because Γ is analytic and nonzero near $\Gamma(1) = 1$. The product rule gives $(Ph)' = P'h + Ph'$, and $P(-m) = 0$ kills the second term, so no derivative of Γ is needed: only $P'(-m)h(-m) = P'(-m)$ survives. For the same reason only one summand of $P'(\nu) = \sum_{i=1}^m \prod_{j \neq i} (\nu + j)$ survives at $\nu = -m$, the one with $i = m$, since every other summand still contains the factor $\nu + m$. Hence

$$\begin{aligned} \left. \frac{d}{d\nu} \frac{1}{\Gamma(\nu + 1)} \right|_{\nu = -m} &= \frac{(-m + 1)(-m + 2) \cdots (-1)}{\Gamma(1)} \\ &= (-1)^{m-1} (m - 1)!, \end{aligned}$$

Small- x behavior of Y_m IV

the last line counting the $m - 1$ negative factors, whose absolute values are $m - 1, m - 2, \dots, 1$. For $m = 1$, the product in the first line is empty and is interpreted as 1. Let $L = \log(x/2)$. The derivative of the $k = 0$ term is

$$\begin{aligned} & \left. \frac{\partial}{\partial \nu} \left[\frac{(x/2)^\nu}{\Gamma(\nu + 1)} \right] \right|_{\nu=-m} \\ &= \left(\frac{x}{2} \right)^{-m} \left[L \frac{1}{\Gamma(\nu + 1)} \Big|_{\nu=-m} + \frac{d}{d\nu} \frac{1}{\Gamma(\nu + 1)} \Big|_{\nu=-m} \right] \\ &= (-1)^{m-1} (m-1)! \left(\frac{x}{2} \right)^{-m}, \end{aligned}$$

because $1/\Gamma(1 - m) = 0$.

The same care is needed for the remaining terms. The ψ -form of the termwise derivative is unusable at $\nu = -m$ when $1 \leq k \leq m - 1$: there $1/\Gamma(k - m + 1) = 0$ while $\psi(k - m + 1)$ has a pole, an indeterminate $0 \cdot \infty$. Repeat instead the rewrite used for $k = 0$. Iterating $\Gamma(z + 1) = z\Gamma(z)$ gives, for $0 \leq k \leq m - 1$,

$$\frac{1}{\Gamma(\nu + k + 1)} = \frac{(\nu + k + 1)(\nu + k + 2) \cdots (\nu + m)}{\Gamma(\nu + m + 1)},$$

Small- x behavior of Y_m

again with the single vanishing factor $\nu + m$ at $\nu = -m$, so the product-rule argument above applies verbatim and yields the finite value

$$\left. \frac{d}{d\nu} \frac{1}{\Gamma(\nu + k + 1)} \right|_{\nu=-m} = (-m+k+1)(-m+k+2) \cdots (-1) = (-1)^{m-k-1} (m-k-1)!.$$

Since $1/\Gamma(k-m+1) = 0$ removes the L contribution, the derivative of the k -th term is

$$\frac{(-1)^k (-1)^{m-k-1} (m-k-1)!}{k!} \left(\frac{x}{2}\right)^{2k-m} = O(x^{2k-m}), \quad 0 \leq k \leq m-1,$$

with no logarithm. For $k \geq m$ the gamma factor is regular, $\Gamma(k-m+1) = (k-m)!$, the ψ -form applies as written, and the derivative of the k -th term is

$$\frac{(-1)^k}{k! (k-m)!} \left(\frac{x}{2}\right)^{2k-m} [L - \psi(k-m+1)] = O(x^{2k-m} |\log x|).$$

Every term with $k \geq 1$ thus carries at least two extra powers of x relative to the $k=0$ term $(-1)^{m-1} (m-1)! (x/2)^{-m}$, and the resulting series converges for each $x > 0$, so the whole tail $\sum_{k \geq 1}$ contributes a relative $O(x^2 |\log x|)$. Hence

$$\left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=-m} = (-1)^{m-1} (m-1)! \left(\frac{x}{2}\right)^{-m} [1 + O(x^2 |\log x|)].$$

Small- x behavior of Y_m VI

Substitution into (6.5.6) gives

$$\begin{aligned} Y_m(x) &= \frac{1}{\pi} \left[O(x^m |\log x|) + (-1)^m (-1)^{m-1} (m-1)! \left(\frac{x}{2}\right)^{-m} [1 + O(x^2 |\log x|)] \right] \\ &= -\frac{(m-1)!}{\pi} \left(\frac{2}{x}\right)^m [1 + o(1)]. \end{aligned}$$

Since $(m-1)! = \Gamma(m)$,

$$Y_m(x) = -\frac{\Gamma(m)}{\pi} \left(\frac{2}{x}\right)^m [1 + o(1)], \quad m = 1, 2, \dots, \quad (6.5.8)$$

as $x \rightarrow 0^+$ with m fixed.

The case $m = 0$ is logarithmic. Setting $m = 0$ in (6.5.6) gives

$$Y_0(x) = \frac{2}{\pi} \left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=0}. \quad (6.5.9)$$

The series and its order derivative converge locally uniformly for ν near zero when $x > 0$ is fixed, so the general-order series may be differentiated term by term. Let

Small- x behavior of Y_m VII

$\psi(z) = \Gamma'(z)/\Gamma(z)$ be the digamma function, let γ be the Euler–Mascheroni constant, and write $L = \log(x/2)$. Then

$$\begin{aligned} \frac{\partial}{\partial \nu} \left[\frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2} \right)^{2k + \nu} \right] \\ = \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2} \right)^{2k + \nu} [L - \psi(k + \nu + 1)]. \end{aligned}$$

At $\nu = 0$, $\Gamma(k + 1) = k!$, so

$$\left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=0} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2} \right)^{2k} [L - \psi(k + 1)].$$

Two facts about ψ are needed, and both come from Chapter 3. First, taking logarithms in the recurrence $\Gamma(z + 1) = z\Gamma(z)$ and differentiating,

$$\psi(z + 1) = \frac{d}{dz} \log \Gamma(z + 1) = \frac{d}{dz} [\log z + \log \Gamma(z)] = \frac{1}{z} + \psi(z).$$

Second, the value at $z = 1$ is where the Euler–Mascheroni constant enters. Chapter 3 established Euler's limit representation

$$\Gamma(z) = \lim_{n \rightarrow \infty} \frac{n^z n!}{z(z + 1) \cdots (z + n)}$$

Small- x behavior of Y_m VIII

together with the definition $\gamma = \lim_{n \rightarrow \infty} (H_n - \log n)$, where $H_n = \sum_{j=1}^n j^{-1}$. Taking logarithms in the limit representation,

$$\log \Gamma(z) = \lim_{n \rightarrow \infty} \left[z \log n + \log n! - \sum_{k=0}^n \log(z+k) \right],$$

and differentiating in z — the convergence is locally uniform away from the nonpositive integers, so the limit and the derivative may be interchanged —

$$\psi(z) = \lim_{n \rightarrow \infty} \left[\log n - \sum_{k=0}^n \frac{1}{z+k} \right].$$

At $z = 1$ the sum is $\sum_{k=0}^n \frac{1}{k+1} = H_{n+1} = H_n + \frac{1}{n+1}$, so

$$\psi(1) = \lim_{n \rightarrow \infty} \left[\log n - H_n - \frac{1}{n+1} \right] = - \lim_{n \rightarrow \infty} (H_n - \log n) = -\gamma.$$

Iterating the recurrence from $z = 1$ now gives

$$\begin{aligned} \psi(k+1) &= \psi(1) + \sum_{j=1}^k \frac{1}{j} \\ &= -\gamma + H_k, \end{aligned}$$

Small- x behavior of Y_m IX

where $H_0 = 0$ and $H_k = \sum_{j=1}^k j^{-1}$. Therefore

$$\begin{aligned}\left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=0} &= (L + \gamma) \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k} \\ &\quad - \sum_{k=1}^{\infty} \frac{(-1)^k H_k}{(k!)^2} \left(\frac{x}{2}\right)^{2k} \\ &= (L + \gamma) J_0(x) - \sum_{k=1}^{\infty} \frac{(-1)^k H_k}{(k!)^2} \left(\frac{x}{2}\right)^{2k}.\end{aligned}$$

As $x \rightarrow 0^+$,

$$J_0(x) = 1 + O(x^2), \quad \sum_{k=1}^{\infty} \frac{(-1)^k H_k}{(k!)^2} \left(\frac{x}{2}\right)^{2k} = O(x^2).$$

Using (6.5.9),

$$\begin{aligned}Y_0(x) &= \frac{2}{\pi} [(L + \gamma)(1 + O(x^2)) + O(x^2)] \\ &= \frac{2}{\pi} (L + \gamma) + O(x^2 |\log x|).\end{aligned}$$

Small- x behavior of $Y_m X$

Thus

$$Y_0(x) = \frac{2}{\pi} \left[\log \left(\frac{x}{2} \right) + \gamma \right] + O(x^2 |\log x|), \quad x \rightarrow 0^+. \quad (6.5.10)$$

Large- x behavior of J_ν and Y_ν I

For fixed order ν , define

$$\chi_\nu(x) = x - \frac{\nu\pi}{2} - \frac{\pi}{4}, \quad A(x) = \sqrt{\frac{2}{\pi x}}.$$

The large- x expansion of J_ν has the leading form

$$J_\nu(x) = A(x) \cos \chi_\nu(x) + O(x^{-3/2}). \quad (6.5.11)$$

This is (6.4.6) for integer order (stationary phase), extended to any fixed real order as noted after that derivation; in particular both ν and $-\nu$ are covered, which is all that is needed below. For $J_{-\nu}$, the phase is

$$\begin{aligned} \chi_{-\nu}(x) &= x + \frac{\nu\pi}{2} - \frac{\pi}{4} \\ &= \chi_\nu(x) + \pi\nu. \end{aligned}$$

Hence

$$J_{-\nu}(x) = A(x) \cos(\chi_\nu(x) + \pi\nu) + O(x^{-3/2}).$$

Large- x behavior of J_ν and Y_ν II

Recall the noninteger-order definition $Y_\nu(x) = \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)}$ from (6.5.4).

For fixed noninteger ν , substituting $J_\nu = A \cos \chi_\nu$ and $J_{-\nu} = A \cos(\chi_\nu + \pi\nu)$ and factoring $A(x)$ out of the numerator gives

$$Y_\nu(x) = A(x) \frac{\cos(\pi\nu) \cos \chi_\nu - \cos(\chi_\nu + \pi\nu)}{\sin(\pi\nu)} + O(x^{-3/2}).$$

Expand the cosine of a sum:

$$\cos(\chi_\nu + \pi\nu) = \cos \chi_\nu \cos(\pi\nu) - \sin \chi_\nu \sin(\pi\nu).$$

Therefore the numerator simplifies term by term:

$$\begin{aligned} & \cos(\pi\nu) \cos \chi_\nu - \cos(\chi_\nu + \pi\nu) \\ &= \cos(\pi\nu) \cos \chi_\nu - \cos \chi_\nu \cos(\pi\nu) + \sin \chi_\nu \sin(\pi\nu) \\ &= \sin \chi_\nu \sin(\pi\nu). \end{aligned}$$

After division by $\sin(\pi\nu)$,

$$Y_\nu(x) = A(x) \sin \chi_\nu(x) + O(x^{-3/2})$$

for fixed noninteger ν .

For integer order, use the order-derivative formula rather than divide by a vanishing sine. At $\nu = m$ the quotient (6.5.4) is of the form $0/0$: $\sin(\pi\nu) \rightarrow 0$, and since $J_{-m} = (-1)^m J_m$ and $\cos(\pi m) = (-1)^m$ the numerator also vanishes, so we cannot

Large- x behavior of J_ν and Y_ν III

substitute directly. Instead use the l'Hôpital result (6.5.6),

$Y_m(x) = \frac{1}{\pi} [\partial_\nu J_\nu|_{\nu=m} + (-1)^m \partial_\nu J_\nu|_{\nu=-m}]$, and insert the large- x form of $\partial_\nu J_\nu$.

Differentiating an asymptotic estimate with respect to a parameter is a step that has to be earned: a bound that is uniform in ν does not by itself control the ν -derivative. For example $R_0(\nu, x) = x^{-3/2} \sin(\nu x^2)$ obeys $|R_0(\nu, x)| \leq x^{-3/2}$ for every real ν , yet $\partial_\nu R_0 = x^{1/2} \cos(\nu x^2)$ grows with x . What licenses the step here is analyticity in the order — the property already used to form the order derivatives in (6.5.6) — together with the fact recorded when (6.5.11) was extended off the integers: the remainder there is uniform for ν in bounded subsets of the complex plane. Write that remainder as

$$R(\nu, x) := J_\nu(x) - A(x) \cos \chi_\nu(x).$$

For fixed $x > 0$ the series for $J_\nu(x)$ is entire in ν , and $A(x) \cos \chi_\nu(x)$ is a cosine of a linear function of ν , hence also entire; so $R(\cdot, x)$ is entire in ν , and on the two circles $|\zeta - m| = 1$ and $|\zeta + m| = 1$ the uniform bound supplies constants C_m and x_0 with

$$|R(\zeta, x)| \leq C_m x^{-3/2} \quad \text{for } |\zeta - m| = 1 \text{ or } |\zeta + m| = 1, \quad x \geq x_0.$$

Cauchy's integral formula for the first derivative, on the circle $|\zeta - m| = 1$, converts that bound on R into a bound on $\partial_\nu R$:

$$\left. \frac{\partial R}{\partial \nu} \right|_{\nu=m} = \frac{1}{2\pi i} \oint_{|\zeta-m|=1} \frac{R(\zeta, x)}{(\zeta - m)^2} d\zeta,$$

Large- x behavior of J_ν and Y_ν IV

and bounding the integrand by its maximum modulus $C_m x^{-3/2}/1^2$ and the contour length by 2π ,

$$\left| \frac{\partial R}{\partial \nu} \right|_{\nu=m} \leq \frac{1}{2\pi} \cdot 2\pi \cdot C_m x^{-3/2} = C_m x^{-3/2},$$

with the identical estimate on $|\zeta + m| = 1$ bounding $\partial_\nu R|_{\nu=-m}$. The remainder in (6.5.11) may therefore be differentiated in ν at $\nu = \pm m$ without changing its order, which is exactly the term-by-term differentiation used below. Since

$$\frac{\partial \chi_\nu}{\partial \nu} = -\frac{\pi}{2},$$

we have

$$\begin{aligned} \frac{\partial}{\partial \nu} [A(x) \cos \chi_\nu(x)] &= A(x) [-\sin \chi_\nu(x)] \left(-\frac{\pi}{2} \right) \\ &= \frac{\pi}{2} A(x) \sin \chi_\nu(x). \end{aligned}$$

Thus

$$\begin{aligned} \left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=m} &= \frac{\pi}{2} A(x) \sin \chi_m(x) + O(x^{-3/2}), \\ \left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=-m} &= \frac{\pi}{2} A(x) \sin \chi_{-m}(x) + O(x^{-3/2}). \end{aligned}$$

Large- x behavior of J_ν and Y_ν

Because

$$\begin{aligned}\chi_{-m}(x) &= \chi_m(x) + m\pi, \\ \sin \chi_{-m}(x) &= (-1)^m \sin \chi_m(x),\end{aligned}$$

substitution into (6.5.6) gives

$$\begin{aligned}Y_m(x) &= \frac{1}{\pi} \left[\frac{\pi}{2} A \sin \chi_m + (-1)^m \frac{\pi}{2} A \sin \chi_{-m} \right] + O(x^{-3/2}) \\ &= \frac{1}{\pi} \left[\frac{\pi}{2} A \sin \chi_m + \frac{\pi}{2} A \sin \chi_m \right] + O(x^{-3/2}) \\ &= A(x) \sin \chi_m(x) + O(x^{-3/2}).\end{aligned}$$

Consequently, for fixed real order, including integer order,

$$Y_\nu(x) = \sqrt{\frac{2}{\pi x}} \sin \chi_\nu(x) + O(x^{-3/2}), \quad x \rightarrow +\infty. \quad (6.5.12)$$

Thus J_m and Y_m have the same leading amplitude and phases separated by $\pi/2$.

Table of contents

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation
- 4 Asymptotics
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series**
- 7 Summary

Hankel functions: Bessel functions of the third kind I

Define

$$H_m^{(1)}(x) = J_m(x) + iY_m(x), \quad (6.6.1)$$

$$H_m^{(2)}(x) = J_m(x) - iY_m(x). \quad (6.6.2)$$

Let

$$\chi_m(x) = x - \frac{m\pi}{2} - \frac{\pi}{4}.$$

Using (6.5.11) and (6.5.12),

$$\begin{aligned} H_m^{(1)}(x) &= \sqrt{\frac{2}{\pi x}} (\cos \chi_m + i \sin \chi_m) + O(x^{-3/2}) \\ &= \sqrt{\frac{2}{\pi x}} e^{i\chi_m(x)} + O(x^{-3/2}), \end{aligned}$$

$$\begin{aligned} H_m^{(2)}(x) &= \sqrt{\frac{2}{\pi x}} (\cos \chi_m - i \sin \chi_m) + O(x^{-3/2}) \\ &= \sqrt{\frac{2}{\pi x}} e^{-i\chi_m(x)} + O(x^{-3/2}). \end{aligned}$$

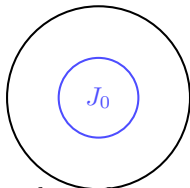
With $k > 0$ and time dependence $e^{-i\omega t}$,

$$H_m^{(1)}(kr)e^{-i\omega t} \propto r^{-1/2} e^{i(kr - \omega t)}$$

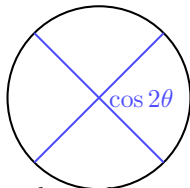
Hankel functions: Bessel functions of the third kind II

is outgoing, whereas $H_m^{(2)}$ is incoming. Since Y_m is singular at the origin, both Hankel functions are singular there; they are far-field wave functions, not replacements for regular J_m modes near $r = 0$.

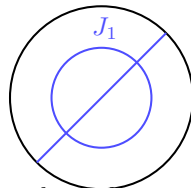
Discrete radial modes on the unit disk



angular order $m = 0$
radial mode 2



angular order $m = 2$
radial mode 1
 θ is the polar angle



angular order $m = 1$
radial mode 2

The boundary condition $J_m(\alpha_{m,j}) = 0$ places a radial zero at the clamped rim $r = 1$. Distinct zeros give distinct Sturm–Liouville eigenvalues $\lambda_j = \alpha_{m,j}^2$. On the infinite half-line the discrete Fourier–Bessel sum becomes a Hankel integral transform (not developed here).

Scaling Bessel's equation and obtaining Sturm–Liouville form I

Step 1: rescale Bessel's equation. Let $z = \alpha s$ and define

$$y(s) = J_m(\alpha s) = J_m(z).$$

By the chain rule,

$$y'(s) = \alpha J'_m(z), \quad y''(s) = \alpha^2 J''_m(z).$$

Bessel's equation in the variable z is

$$z^2 J''_m(z) + z J'_m(z) + (z^2 - m^2) J_m(z) = 0.$$

Substitute $z = \alpha s$ and the chain-rule expressions $J'_m = y'/\alpha$, $J''_m = y''/\alpha^2$:

$$\begin{aligned} 0 &= (\alpha s)^2 \frac{y''}{\alpha^2} + (\alpha s) \frac{y'}{\alpha} + ((\alpha s)^2 - m^2) y \\ &= s^2 y'' + s y' + (\alpha^2 s^2 - m^2) y. \end{aligned}$$

Thus

$$s^2 y''(s) + s y'(s) + (\alpha^2 s^2 - m^2) y(s) = 0. \quad (6.6.3)$$

Divide by $s > 0$:

$$s y'' + y' + \left(\alpha^2 s - \frac{m^2}{s} \right) y = 0.$$

Scaling Bessel's equation and obtaining Sturm–Liouville form II

Since

$$\frac{d}{ds}(sy') = sy'' + y',$$

we obtain

$$\frac{d}{ds}(sy') - \frac{m^2}{s}y = -\alpha^2 sy.$$

Therefore the Sturm–Liouville form is

$$\frac{d}{ds} \left(s \frac{dy}{ds} \right) - \frac{m^2}{s} y = -\alpha^2 s y. \quad (6.6.4)$$

In the convention

$$(py')' + qy = -\lambda wy,$$

we identify

$$p(s) = s, \quad q(s) = -\frac{m^2}{s}, \quad \lambda = \alpha^2, \quad w(s) = s.$$

Hence the natural inner product is (linear in the first slot, matching Chapter 5)

$$\langle u, v \rangle = \int_0^1 u(s) \overline{v(s)} s \, ds. \quad (6.6.5)$$

Scaling Bessel's equation and obtaining Sturm–Liouville form III

For the real Bessel modes used below, the complex conjugate may be omitted. Write $y_j(s) := J_m(\alpha_{m,j}s)$ for two disk modes y_j, y_k . Self-adjointness of the operator $\mathcal{L}y = (sy')' - \frac{m^2}{s}y$ gives Lagrange's identity

$$y_j \mathcal{L}y_k - y_k \mathcal{L}y_j = \frac{d}{ds} [s(y_j y'_k - y_k y'_j)],$$

so integrating over $[0, 1]$ leaves the boundary term $[s(y_j y'_k - y_k y'_j)]_0^1$ (the same Wronskian term evaluated in the next frame). At the lower endpoint $s = 0$, regularity removes this term. For $m \geq 1$,

$$J_m(\alpha s) = O(s^m), \quad \frac{d}{ds} J_m(\alpha s) = O(s^{m-1}),$$

so

$$s(y_j y'_k - y_k y'_j) = O(s^{2m}) \rightarrow 0.$$

For $m = 0$, $J_0(\alpha s) = 1 + O(s^2)$ and its s -derivative is $O(s)$, so the same boundary term is $O(s^2) \rightarrow 0$. Singular Y_m modes do not satisfy the regular endpoint condition.

Fourier–Bessel orthogonality: distinct eigenvalues I

Step 1: compare two eigenvalues. For positive parameters a and b , set

$$y_a(x) = J_m(ax), \quad y_b(x) = J_m(bx).$$

They satisfy

$$(xy'_a)' + \left(a^2x - \frac{m^2}{x}\right)y_a = 0,$$

$$(xy'_b)' + \left(b^2x - \frac{m^2}{x}\right)y_b = 0.$$

Multiply the first equation by y_b , the second by y_a , and subtract:

$$y_b(xy'_a)' - y_a(xy'_b)' + (a^2 - b^2)xy_ay_b = 0.$$

The first two terms form a total derivative. Applying the product rule to every factor,

$$\begin{aligned} \frac{d}{dx} [x(y_by'_a - y_ay'_b)] &= y_by'_a - y_ay'_b \\ &\quad + x(y'_by'_a + y_by''_a - y'_ay'_b - y_ay''_b) \\ &= y_by'_a - y_ay'_b + x(y_by''_a - y_ay''_b) \\ &= y_b(y'_a + xy''_a) - y_a(y'_b + xy''_b) \\ &= y_b(xy'_a)' - y_a(xy'_b)'. \end{aligned}$$

Fourier–Bessel orthogonality: distinct eigenvalues II

Therefore

$$\frac{d}{dx} [x(y_b y'_a - y_a y'_b)] + (a^2 - b^2) x y_a y_b = 0.$$

Integrate from 0 to 1:

$$[x(y_b y'_a - y_a y'_b)]_0^1 + (a^2 - b^2) \int_0^1 x y_a(x) y_b(x) dx = 0.$$

The lower boundary is zero by regularity. At $x = 1$,

$$y_a(1) = J_m(a), \quad y'_a(1) = a J'_m(a), \quad y_b(1) = J_m(b), \quad y'_b(1) = b J'_m(b).$$

Hence the boundary evaluation is

$$\begin{aligned} [x(y_b y'_a - y_a y'_b)]_0^1 &= 1 \cdot (J_m(b) \cdot a J'_m(a) - J_m(a) \cdot b J'_m(b)) \\ &= a J_m(b) J'_m(a) - b J_m(a) J'_m(b). \end{aligned}$$

Solving for the integral gives the cross-product formula

$$\int_0^1 x J_m(ax) J_m(bx) dx = \frac{b J_m(a) J'_m(b) - a J_m(b) J'_m(a)}{a^2 - b^2}, \quad a \neq b. \quad (6.6.6)$$

(The numerator is the negative of the boundary evaluation above, which cancels the overall minus sign from moving $(a^2 - b^2) \int$ to the other side.)

Fourier–Bessel orthogonality: distinct eigenvalues III

Step 2: impose the zeros at the rim. Now let $a = \alpha_{m,j}$ and $b = \alpha_{m,k}$ be distinct positive zeros of J_m . Then

$$J_m(a) = J_m(b) = 0,$$

so the numerator in (6.6.6) is zero. Since $a^2 - b^2 \neq 0$,

$$\int_0^1 x J_m(\alpha_{m,j} x) J_m(\alpha_{m,k} x) dx = 0, \quad j \neq k. \quad (6.6.7)$$

Fourier–Bessel normalization: the diagonal case I

Existence of the zeros. For fixed integer $m \geq 0$, the large- x cosine law (6.4.6) oscillates with amplitude $\sim x^{-1/2}$ and phase advancing like x , so J_m changes sign infinitely often as $x \rightarrow +\infty$. Hence J_m has infinitely many positive zeros $\alpha_{m,1} < \alpha_{m,2} < \cdots$ (alternatively: Sturm comparison of Bessel's equation in normal form against $y'' + y = 0$ on large intervals yields the same conclusion). We index them increasingly and write $\alpha_{m,j}$ for the j th.

Every positive zero $a = \alpha_{m,j}$ is simple, that is, $J'_m(a) \neq 0$. Nothing new is needed to see this: with the Wronskian convention $W[u, v] = uv' - u'v$ fixed earlier, the normalization (6.5.3) reads

$$J_m(x)Y'_m(x) - J'_m(x)Y_m(x) = \frac{2}{\pi x}, \quad x > 0,$$

and at any point $a > 0$ the second solution Y_m and its derivative are finite, since Y_m is singular only at the origin. Suppose both $J_m(a) = 0$ and $J'_m(a) = 0$. Evaluating the identity at $x = a$ makes each product vanish separately,

$$\underbrace{J_m(a)}_{=0} Y'_m(a) - \underbrace{J'_m(a)}_{=0} Y_m(a) = 0,$$

while the right-hand side equals $2/(\pi a) \neq 0$ because a is finite and positive. The two sides disagree, so J_m and J'_m cannot vanish at the same positive point:

$$J'_m(\alpha_{m,j}) \neq 0 \quad \text{for every } j \geq 1.$$

Fourier–Bessel normalization: the diagonal case II

This is exactly the statement that a positive zero of J_m is a simple zero. It is what keeps the normalization computed at the end of this frame strictly positive, so that the coefficient formula of the next frame divides by a nonzero number.

To compute the norm at such a zero, take the limit $b \rightarrow a$ in (6.6.6). The integral passes to the limit by continuity of the integrand on the compact interval $0 \leq x \leq 1$. Write the numerator and denominator as functions of b with a fixed and $J_m(a) = 0$:

$$\begin{aligned}N(b) &:= bJ_m(a)J'_m(b) - aJ_m(b)J'_m(a) = -aJ_m(b)J'_m(a), \\D(b) &:= a^2 - b^2.\end{aligned}$$

Both $N(a)$ and $D(a)$ vanish, so the limit is a 0/0 form. Apply L'Hôpital: differentiate with respect to b ,

$$\begin{aligned}N'(b) &= -aJ'_m(b)J'_m(a), \\D'(b) &= -2b.\end{aligned}$$

Therefore

$$\lim_{b \rightarrow a} \frac{N(b)}{D(b)} = \lim_{b \rightarrow a} \frac{N'(b)}{D'(b)} = \frac{-aJ'_m(a)J'_m(a)}{-2a} = \frac{a[J'_m(a)]^2}{2a} = \frac{1}{2} [J'_m(a)]^2,$$

Fourier–Bessel normalization: the diagonal case III

where we used $a > 0$. Equivalently, with $b = a + \varepsilon$, Taylor expansion gives the same result with every algebra line visible:

$$\begin{aligned}J_m(b) &= J_m(a) + J'_m(a)\varepsilon + O(\varepsilon^2) = J'_m(a)\varepsilon + O(\varepsilon^2), \\a^2 - b^2 &= a^2 - (a + \varepsilon)^2 = -2a\varepsilon - \varepsilon^2,\end{aligned}$$

so

$$\begin{aligned}\frac{N(b)}{D(b)} &= \frac{-a[J'_m(a)\varepsilon + O(\varepsilon^2)]J'_m(a)}{-2a\varepsilon - \varepsilon^2} = \frac{-a[J'_m(a)]^2\varepsilon + O(\varepsilon^2)}{-2a\varepsilon - \varepsilon^2} \\&= \frac{-a[J'_m(a)]^2 + O(\varepsilon)}{-2a - \varepsilon} \xrightarrow{\varepsilon \rightarrow 0} \frac{a[J'_m(a)]^2}{2a} = \frac{1}{2}[J'_m(a)]^2.\end{aligned}$$

Thus

$$\int_0^1 x J_m^2(ax) dx = \frac{1}{2}[J'_m(a)]^2 \quad \text{at a positive zero } a \text{ of } J_m. \quad (6.6.8)$$

At a zero of J_m , the derivative recurrence

$$J'_m(a) = \frac{m}{a}J_m(a) - J_{m+1}(a)$$

reduces to

$$J'_m(a) = -J_{m+1}(a).$$

Fourier–Bessel normalization: the diagonal case IV

Squaring removes the sign, and hence

$$\int_0^1 x J_m^2(\alpha_{m,j} x) dx = \frac{1}{2} J_{m+1}^2(\alpha_{m,j}). \quad (6.6.9)$$

Combining the diagonal and off-diagonal cases gives

$$\int_0^1 J_m(\alpha_{m,j} x) J_m(\alpha_{m,k} x) x dx = \frac{1}{2} J_{m+1}^2(\alpha_{m,j}) \delta_{jk}. \quad (6.6.10)$$

Fourier–Bessel coefficients I

Assume a radial function of angular order m has the disk expansion

$$f(x) = \sum_{j=1}^{\infty} a_j J_m(\alpha_{m,j}x), \quad 0 < x < 1, \quad (6.6.11)$$

where, exactly as for the sine series in Chapter 5, the equality is understood in the weighted space $L^2((0, 1), x dx)$ attached to the inner product (6.6.5). Writing the partial sums and the weighted norm

$$S_N(x) = \sum_{j=1}^N a_j J_m(\alpha_{m,j}x), \quad \|g\|^2 = \int_0^1 |g(x)|^2 x dx,$$

the hypothesis is $\|f - S_N\| \rightarrow 0$ as $N \rightarrow \infty$.

Take the inner product with the k th mode. The test mode has finite norm, because (6.6.9) evaluates it exactly:

$$\|J_m(\alpha_{m,k} \cdot)\|^2 = \int_0^1 J_m^2(\alpha_{m,k}x) x dx = \frac{1}{2} J_{m+1}^2(\alpha_{m,k}) < \infty.$$

Hence the Cauchy–Schwarz inequality in $L^2((0, 1), x dx)$ controls the error committed by replacing f with S_N under the integral sign:

$$\left| \int_0^1 (f(x) - S_N(x)) J_m(\alpha_{m,k}x) x dx \right| \leq \|f - S_N\| \|J_m(\alpha_{m,k} \cdot)\| \xrightarrow{N \rightarrow \infty} 0.$$

Fourier–Bessel coefficients II

So the integral against f is the limit of the integrals against S_N , and for each fixed N the sum is finite, where only linearity of the integral is used:

$$\begin{aligned}\int_0^1 f(x) J_m(\alpha_{m,k} x) x \, dx &= \lim_{N \rightarrow \infty} \int_0^1 S_N(x) J_m(\alpha_{m,k} x) x \, dx \\ &= \lim_{N \rightarrow \infty} \sum_{j=1}^N a_j \int_0^1 J_m(\alpha_{m,j} x) J_m(\alpha_{m,k} x) x \, dx \\ &= \lim_{N \rightarrow \infty} \sum_{j=1}^N a_j \frac{1}{2} J_{m+1}^2(\alpha_{m,j}) \delta_{jk} = \frac{1}{2} a_k J_{m+1}^2(\alpha_{m,k}),\end{aligned}$$

where (6.6.10) was used in the third line, and the last equality holds because as soon as $N \geq k$ the Kronecker delta leaves the single surviving term $j = k$, so the sequence of partial sums is eventually constant. Solving for a_k gives

$$a_k = \frac{2}{J_{m+1}^2(\alpha_{m,k})} \int_0^1 f(r) J_m(\alpha_{m,k} r) r \, dr. \quad (6.6.12)$$

This is the discrete Fourier–Bessel coefficient formula on the unit disk with weight r . Completeness of $\{J_m(\alpha_{m,j} x)\}_{j \geq 1}$ in the weighted space $L^2((0, 1), x \, dx)$ is the standard Sturm–Liouville spectral theorem for the singular SL problem (6.6.4) with

Fourier–Bessel coefficients III

the regular endpoint condition at $s = 0$ and $y(1) = 0$; we cite it as background and do not re-prove the general SL expansion theorem here.

Table of contents

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation
- 4 Asymptotics
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series
- 7 Summary**

Formula sheet: J_m I

For integer $m \geq 0$,

$$x^2 J_m''(x) + x J_m'(x) + (x^2 - m^2) J_m(x) = 0,$$

$$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(m+k)!} \left(\frac{x}{2}\right)^{2k+m},$$

$$J_{-m}(x) = (-1)^m J_m(x).$$

The generating function, valid for $t \neq 0$, is

$$e^{\frac{x}{2}(t-t^{-1})} = \sum_{m=-\infty}^{\infty} J_m(x) t^m.$$

The recurrences are

$$J_{m-1}(x) + J_{m+1}(x) = \frac{2m}{x} J_m(x), \quad x \neq 0,$$

$$2J_m'(x) = J_{m-1}(x) - J_{m+1}(x),$$

$$J_m'(x) = J_{m-1}(x) - \frac{m}{x} J_m(x), \quad x \neq 0,$$

$$J_m'(x) = \frac{m}{x} J_m(x) - J_{m+1}(x), \quad x \neq 0.$$

Formula sheet: J_m II

The preferred integral representations are

$$J_m(x) = \frac{1}{\pi} \int_0^\pi \cos(m\theta - x \sin \theta) d\theta, \quad m \geq 0,$$

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta, \quad n \in \mathbb{Z},$$

with the optional contour form

$$J_n(x) = \frac{1}{2\pi i} \oint_C e^{\frac{x}{2}(t-t^{-1})} t^{-n-1} dt, \quad n \in \mathbb{Z}.$$

For fixed m ,

$$J_m(x) = \frac{1}{m!} \left(\frac{x}{2}\right)^m \left[1 - \frac{x^2}{4(m+1)} + O(x^4)\right], \quad x \rightarrow 0,$$

$$J_m(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}), \quad x \rightarrow +\infty.$$

Formula sheet: Y_m , Hankel, Fourier–Bessel I

For noninteger ν ,

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)},$$

and for integer m ,

$$Y_m(x) = \frac{1}{\pi} \left[\left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=m} + (-1)^m \left. \frac{\partial J_\nu(x)}{\partial \nu} \right|_{\nu=-m} \right].$$

The normalization is

$$W[J_m, Y_m] = \frac{2}{\pi x}.$$

For fixed integer $m \geq 1$,

$$Y_m(x) = -\frac{\Gamma(m)}{\pi} \left(\frac{2}{x}\right)^m [1 + o(1)], \quad x \rightarrow 0^+,$$

whereas

$$Y_0(x) = \frac{2}{\pi} \left[\log\left(\frac{x}{2}\right) + \gamma \right] + O(x^2 |\log x|).$$

For fixed m and $x \rightarrow +\infty$,

$$Y_m(x) = \sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}).$$

Formula sheet: Y_m , Hankel, Fourier–Bessel II

The Hankel functions are

$$\begin{aligned}H_m^{(1)} &= J_m + iY_m, & H_m^{(1)}(x) &= \sqrt{\frac{2}{\pi x}} e^{i\chi_m(x)} + O(x^{-3/2}), \\H_m^{(2)} &= J_m - iY_m, & H_m^{(2)}(x) &= \sqrt{\frac{2}{\pi x}} e^{-i\chi_m(x)} + O(x^{-3/2}),\end{aligned}$$

with $\chi_m(x) = x - m\pi/2 - \pi/4$.

For the positive zeros $\alpha_{m,j}$ of J_m ,

$$\int_0^1 J_m(\alpha_{m,j}x) J_m(\alpha_{m,k}x) x \, dx = \frac{1}{2} J_{m+1}^2(\alpha_{m,j}) \delta_{jk},$$

If a radial function of angular order m is expanded on the disk as

$$f(r) = \sum_{j=1}^{\infty} a_j J_m(\alpha_{m,j}r), \quad 0 < r < 1,$$

then multiplying by $J_m(\alpha_{m,k}r)r$, integrating over $(0, 1)$ and using the orthogonality relation above (only the term $j = k$ survives) gives the coefficients

$$a_j = \frac{2}{J_{m+1}^2(\alpha_{m,j})} \int_0^1 f(r) J_m(\alpha_{m,j}r) r \, dr,$$

as derived in (6.6.11)–(6.6.12).